

Drift-Resistant Heading Estimation for Smartphone-Based Indoor Positioning via Adaptive Calibration Using Wi-Fi Fingerprinting and Magnetic Stability

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Abstract—Accurate heading estimation is essential for improving position accuracy and enabling intuitive navigation by indicating the correct direction to the destination, yet it remains challenging on smartphones because of short-term compass bias, long-term gyroscope drift, and unconstrained user motion. Previously proposed external infrastructure-dependent methods demonstrate improved heading estimation; however, such reliance faces scalability issues. Conversely, pervasively available Wi-Fi fingerprinting approaches primarily focus on position estimates, overlooking opportunities for heading calibration through temporal continuity and motion-state alignment. This article utilizes this alignment with the fingerprinting sequence to infer heading calibration. The approach evaluates the suitability of fingerprint positions for calibrating heading using spatial variance and signal strength, incorporating an adaptive thresholding mechanism driven by continuous self-learning to dynamically refine the qualification criteria. A multilayer validation scheme further improves reliability by cross-correlating displacement-based step estimates and compass-derived headings with corresponding estimates from qualified fingerprinting solutions. To complement this, we propose exploiting quasi-static magnetic field intervals (QSMFIs) for reliable heading calibration. Unlike current QSMFI techniques designed for foot-mounted systems with fixed detection parameters, our method introduces a smartphone-oriented method tailored for unrestricted handheld use. A self-attention mechanism is employed to reject faulty QSMFIs, dynamically adjusting window sizes and detection thresholds based on historical magnetic stability, allowing detection across diverse environments and motion contexts. These calibrated headings are integrated within a loosely coupled extended Kalman filter (EKF) that fuses dead reckoning and Wi-Fi fingerprinting for robust position estimation. Real-world experimental results demonstrate that the

integrated approach significantly outperforms in reducing drift and enhancing positioning robustness.

Index Terms—Heading calibration, indoor positioning systems (IPSS), inertial sensors, magnetic stability detection, pedestrian dead reckoning (PDR), quasi-static magnetic field (QSMF), sensor fusion, smartphone-based navigation, Wi-Fi fingerprinting.

I. INTRODUCTION

INDOOR positioning systems (IPSSs) have become indispensable in various applications, including navigation, emergency response, asset tracking, and location-based services [1]. However, achieving reliable indoor positioning remains challenging due to the unavailability or degradation of global navigation satellite system (GNSS) signals in indoor environments, which suffer from multipath reflections, signal attenuation, and complete signal blockage indoors [2]. These limitations necessitate alternative approaches for robust and accurate indoor positioning. Among these alternatives, infrastructure-based systems offer better accuracy but face scalability limitations due to deployment requirements [3], [4]. In contrast, methods based on embedded smartphone sensors and signal of opportunity, particularly inertial, magnetic sensors, and Wi-Fi-based received signal strength (RSS), offer a widely accessible and scalable solution, making them a preferred choice for mass-market indoor positioning [5].

User displacement and heading are the two fundamental components required for continuously updating the user's position [6]. Displacement can be estimated via velocity integration or gait cycle-based step detection [7]. While over- or under-estimation errors often accompany displacement estimation, these errors are generally limited compared to those of heading estimation [8]. Heading estimation presents more significant challenges due to its higher sensitivity to sensor bias, unconstrained motion challenges, and environmental factors [9]. Reliable heading estimation severely influences the user's trajectory and affects overall position estimation accuracy. Regardless of its role in position estimation, accurate heading plays a key role in navigation applications for guiding the user by reorienting their current direction by displaying the correct directional arrow toward the destination, thereby supporting intuitive and effective guidance.

Embedded inertial sensors, including accelerometers and gyroscopes, along with magnetic sensors, have emerged as

Received 20 August 2025; revised 31 December 2025; accepted 20 January 2026. Date of publication 10 February 2026; date of current version 24 February 2026. This work was supported in part by Hong Kong Research Grants Council (RGC) General Research Fund under Grant 15229622, in part by The Hong Kong Polytechnic University Postdoctoral Matching Fund Scheme, and in part by Shanghai Sci-Tech Co-Research Program under Project 25HB2703600. The Associate Editor coordinating the review process was Dr. Yulong Huang. (Corresponding author: Wu Chen.)

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Digital Object Identifier 10.1109/TIM.2026.3661693

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ubiquitous, scalable sources for heading determination due to their widespread availability in off-the-shelf smartphones. However, despite providing absolute orientation references, the magnetic compass is highly susceptible to bias and severe disturbances in indoor environments, resulting from metallic structures, electrical equipment, and variable magnetic anomalies [10]. While robust to short-term disturbances, gyroscope-based heading estimation suffers from long-term drift attributed primarily to thermal effects and sensor noise accumulation [9]. As a result, long-duration trajectories in unconstrained smartphone use are prone to substantial heading errors that directly degrade positioning performance.

These limitations have motivated a range of infrastructure-assisted and algorithmic calibration methods, yet most either rely on external beacons and specific deployment conditions, or target constrained sensor placements such as foot-mounted units. In contrast, studies aiming for widespread adoption have employed Wi-Fi fingerprinting, leveraging the ubiquitous availability of Wi-Fi access points (APs) [11], in combination with pedestrian dead reckoning (PDR). However, these methods primarily use fingerprinting to update position within fusion frameworks, often missing the opportunity to exploit fingerprint sequences for heading correction by neglecting temporal continuity and motion-state alignment [12], [13]. This omission disregards the valuable contribution that fingerprinting can make in enhancing heading estimation, particularly in scenarios where subsequent fingerprinting solutions exhibit low positional accuracy and are ineffective for position correction. In such cases, leveraging high-quality fingerprinting outputs for heading calibration can substantially improve the PDR system's overall heading performance and resilience.

As another ubiquitous source for heading estimation and correction, the magnetic compass is widely used to initialize and periodically bound gyroscope drift. To improve its reliability, prior work has proposed detecting quasi-static magnetic field intervals (QSMFIs), that is, periods of minimal magnetic field variation that can serve as stable absolute heading references and help suppress short-term disturbances [14]. Although several QSMFI detection strategies have been proposed, they are mostly designed for foot-mounted inertial navigation [15], [16], [17], where constrained motion and stable sensor orientation simplify interval detection and use. In contrast, handheld smartphones exhibit highly variable motion and pose, which complicates robust QSMFI identification. Yet during many practical scenarios, such as navigation or short stationary periods, the device can be sufficiently stable to exploit magnetic steadiness for heading calibration. Existing methods, however, remain largely foot-mounted oriented, rely on fixed window sizes and static variance thresholds that ignore motion context and environmental variability, and are predominantly rule-based rather than context-aware or data-driven, with limited systematic evaluation under realistic smartphone conditions.

To address these challenges, this article proposes an adaptive and robust heading calibration framework for smartphone-based IPS. The approach leverages time-series Wi-Fi fingerprinting and adaptive thresholding to qualify heading corrections, supported by motion context and multilevel

validation. In parallel, we introduce a smartphone-oriented QSMFI framework that dynamically adjusts detection parameters based on magnetic stability behavior and cross-sensor consistency. Together, these mechanisms mitigate drift and sensor bias, enhancing the accuracy, scalability, and reliability of heading estimation in unconstrained smartphone usage. Our contributions are summarized as follows.

- 1) We propose leveraging time-series Wi-Fi fingerprinting for heading calibration rather than solely for position updates. The method qualifies fingerprinting solutions using spatial dispersion and signal-strength-based confidence metrics, enhanced by adaptive thresholding through continuous self-learning. A multilayer validation scheme cross-correlates displacement-based step estimates and compass-derived headings with fingerprinting solutions to ensure calibration reliability.
- 2) Unlike existing foot-mounted QSMFI methods, we introduce a smartphone-tailored approach for unconstrained handheld usage. The method employs motion-context-aware detection strategies, dynamically adjusting sliding-window sizes and variance thresholds based on historical magnetic stability behavior. A self-attention mechanism rejects faulty QSMFIs through cross correlation with gyroscope-integrated angular changes, ensuring only high-confidence heading corrections are applied.
- 3) We propose a loosely coupled extended Kalman filter (EKF)-based fusion scheme in which PDR provides continuous state propagation, while quality-gated Wi-Fi fingerprinting solutions together with QSMFI-derived information are used as intermittent pseudo-measurements to adaptively calibrate the heading. The complete system is implemented in a custom Android application that operates entirely on-device without relying on external computational resources, demonstrating its feasibility in real-world settings.

The remainder of this article is organized as follows. Section II reviews the state-of-the-art in Wi-Fi fingerprinting-based and inertial-magnetic/QSMFI-based heading estimation for smartphone indoor positioning. Section III presents the proposed methodology, including the adaptive heading calibration framework based on Wi-Fi fingerprinting and QSMFI detection. Section IV details the experimental setup, evaluation metrics, and results demonstrating the effectiveness of the proposed approach. Finally, the article is concluded with key findings and future research directions.

II. RELATED WORK

Accurate heading estimation is fundamental to smartphone-based IPSs, yet it remains challenging due to sensor limitations and environmental disturbances [18]. Over the past decade, researchers have explored various approaches to mitigate these challenges, including infrastructure-aided methods, sensor fusion techniques, magnetic field stability detection, and Wi-Fi-based calibration strategies. Despite these advances, existing heading estimation methods struggle to balance accuracy, robustness, and scalability in smartphone-based

indoor positioning [19]. The need for calibration strategies that are both infrastructure-free and adaptive to varying motion contexts and environmental conditions remains a critical challenge. This section reviews the state-of-the-art in Wi-Fi fingerprinting-based and inertial-magnetic/QSMFI-based heading estimation for smartphone indoor positioning. Through this review, we identify critical gaps that motivate the proposed adaptive heading calibration framework.

A. Leveraging Wi-Fi Fingerprinting for Heading Calibration Beyond Positioning

Infrastructure-aided approaches leverage external references such as Bluetooth low energy (BLE) beacons [20] or ultra-wideband (UWB) [21], [22] to provide periodic heading corrections. While effective, these methods face scalability limitations due to infrastructure deployment requirements, making them less suitable for widespread adoption [23]. Wi-Fi fingerprinting is a scalable, infrastructure-free approach for indoor positioning that leverages the ubiquitous availability of APs [24]. Although it is mainly used for position estimation, the underlying temporal and spatial structure can also support heading calibration, a capability that remains insufficiently explored. Recent studies have exploited RSS gradients and spatial signal-strength patterns to support trajectory tracking and direction-aware inference [25], [26], and have further used time-series fingerprint sequences or recurrent models to recover user paths and walking direction from radio measurements [27], [28]. Yet they often overlook how signal variability and environmental dynamics affect estimate stability and thus require adaptation. At the same time, noisy RSS and sparse AP coverage demand strict quality control, but existing methods generally lack robust qualification mechanisms to assess when Wi-Fi-derived headings are trustworthy. This motivates adaptive schemes that determine the suitability of Wi-Fi fingerprints for heading calibration using spatial dispersion, signal-strength confidence, and cross-validation with other sensor modalities.

B. Heading Estimation With Inertial, Magnetic, and QSMFI Approaches

Heading estimation for pedestrian navigation relies primarily on inertial and magnetic sensors embedded in smartphones. Compass-based methods provide absolute orientation references but are highly susceptible to magnetic disturbances in indoor environments. Gyroscope-based methods offer short-term stability but suffer from long-term drift due to bias accumulation. Hybrid approaches attempt to combine these complementary characteristics, while infrastructure-aided methods leverage external references to improve accuracy.

The magnetic compass provides absolute heading measurements by sensing the Earth's magnetic field. However, indoor environments introduce severe magnetic anomalies caused by metallic structures, electrical equipment, and ferromagnetic materials [10]. These disturbances can cause heading errors exceeding 30°, rendering compass measurements unreliable for continuous navigation [29]. Various calibration techniques,

including hard-iron and soft-iron correction, have been proposed to mitigate systematic biases, but they cannot address transient disturbances caused by environmental factors.

Gyroscope integration offers an alternative by computing heading changes through angular velocity integration. While robust to magnetic disturbances, gyroscopes suffer from bias drift that accumulates over time, leading to unbounded heading errors. Thermal effects, manufacturing imperfections, and sensor noise contribute to this drift, necessitating periodic calibration. Zero-velocity updates (ZUPTs) and zero-angular-rate updates (ZARUs) have been successfully applied in foot-mounted systems to reset accumulated errors during stance phases, but these techniques are less effective for handheld smartphones due to unconstrained motion patterns.

Hybrid methods combine compass and gyroscope measurements using complementary filtering or Kalman filtering to leverage their respective strengths. Mahony and Madgwick filters are widely used orientation estimation algorithms that fuse accelerometer, gyroscope, and magnetometer data to provide robust heading estimates. However, these filters still rely on compass measurements for absolute reference, making them vulnerable to magnetic disturbances. QSMFI detection identifies periods of minimal magnetic field variation, during which compass measurements can be considered reliable for heading calibration. The theoretical foundation of QSMFI-based heading estimation was established by Raab [14], who proposed using dipole field models to characterize magnetic behavior for positioning and orientation. While effective in controlled scenarios, QSMFI detection faces significant challenges when applied to unconstrained smartphone usage, where device pose and motion dynamics vary unpredictably. Most existing QSMFI detection methods are designed for foot-mounted inertial navigation systems (INSs), where motion constraints simplify detection. Zhang et al. [15] proposed an improved quasi-static field (QSF) method integrated with a multistage compass filter and iterative map matching for drift correction in foot-mounted IMUs. Their approach effectively suppresses long-term trajectory drift but assumes stable sensor orientation and relies on preexisting maps, limiting its applicability to dynamic smartphone contexts. Ma et al. [16] developed a heading error estimation approach based on an improved QSMFI detection scheme using a generalized likelihood ratio test (GLRT). This method enhances detection accuracy by combining magnetic field stability with heading rate variance. However, the framework employs static detection parameters and assumes straight-line walking, limiting its generalizability to unconstrained mobile device usage. Ma et al. [17] extended their earlier work by proposing an improved QSMF (iQSMF) method with magnetometer calibration and heading correction via heading-gyro deviation. Despite better detection reliability, the method still employs fixed-size sliding windows and static thresholds, restricting its adaptability to varying magnetic conditions and dynamic user motion.

Lee et al. [30] proposed a QSF-aided gyro bias estimation technique that leverages zero-angular-rate detection to refine bias estimation during walking. This work demonstrates that QSMFI can be useful even in motion, provided that the user undergoes periodic stabilization. However, the method

TABLE I
COMPARISON OF QSMFI DETECTION METHODS IN LITERATURE

Study	Platform and Application	Main Contribution	Use of QSMF	Adaptivity
Zhang et al. [15]	Foot-mounted IMU PDR with infrastructure-free positioning.	Proposes improved QSF method with a compass filter and iterative 2D map matching for PDR drift correction.	Enhanced QSF with magnetic preprocessing; integrated into EKF and refined with map matching.	Static thresholds; no motion-adaptive design
Zhang et al. [31]	IMU-based indoor pedestrian tracking using stance-phase detection.	Redesigns QMD to detect hard disturbances and selects EC or HDR algorithms dynamically.	Revised QMD integrated with EC and HDR in a robust AIEZ framework.	Rule-based switching; lacks continuous adaptation
Ma et al. [16]	Foot-mounted navigation in perturbed indoor magnetic environments.	Introduces improved QSF using magnetic and heading rate changes and GLRT theory.	Improved QSF used in stance phases with EKF fusion; detects quasi-static intervals via heading and magnetic magnitude changes.	Static window and threshold with GLRT logic
Lee et al. [30]	Emergency PDR scenarios without long initialization pauses.	Combines QSF and zero angular rate detection for accurate gyro bias estimation during walking.	QSF detection refined with gyro motion comparison for dynamic estimation of biases.	Motion-conditioned, but not sliding-window adaptive
Ma et al. [17]	Robust foot-mounted INS with calibration for complex indoor trajectories.	Generalized heading error suppression using improved QSF with two-axis calibration.	Detects QSF from heading discrepancies and magnetic signal change rate, independent of walking pattern.	Static thresholds; no dynamic tuning
Raab [14]	Subsurface and through-the-earth positioning systems using magnetic sensors.	Lays out theoretical foundations for using quasi-static magnetic fields for 3D orientation and positioning.	Proposes complete model for position/orientation estimation using magnetic dipole fields.	Not applicable; theoretical and static modeling
This work	Smartphone-based indoor navigation with drift-resistant heading correction.	Proposes adaptive QSMFI detection for heading calibration using dynamic window sizing and threshold tuning.	QSMFI detection integrates context-aware thresholds, motion segmentation, and cross-correlation with gyro heading for validation.	Fully adaptive: window size and threshold learned from history and user context

presumes shoe-mounted sensors and structured gait patterns, which are unrealistic for smartphones held in varying orientations. Zhang et al. [31] introduced the advanced IEZ framework, which integrates quasi-static magnetic detection (QMD) with heuristic drift reduction (HDR) and electronic compass (EC) switching. The system intelligently selects between EC and HDR based on disturbance status, improving heading reliability. Yet its switching logic is rule-based and lacks adaptive parameter control based on real-time motion or magnetic dynamics.

As summarized in Table I, existing QSMFI detection methods primarily target foot-mounted systems with constrained motion and rely on static detection parameters. There is a clear gap in the literature regarding adaptive QSMFI detection tailored for smartphone-based systems, where unconstrained device poses and motion dynamics necessitate dynamic parameter adjustment based on historical magnetic stability behavior and cross-sensor validation.

Overall, the literature review reveals three critical gaps that motivate the proposed work. First, existing Wi-Fi fingerprinting-based fusion systems predominantly use fingerprinting for position updates, overlooking opportunities to exploit temporal continuity and motion-state alignment for heading calibration. Second, QSMFI detection methods, while effective for foot-mounted systems with constrained motion, have not been adequately adapted for smartphone-based systems where users exhibit unconstrained device poses and movement dynamics. Most existing QSMFI approaches rely on static detection parameters, limiting their robustness across diverse indoor environments and motion contexts. These gaps underscore the need for an adaptive heading calibration framework specifically designed for smartphone-based indoor positioning, which is the focus of this work.

III. METHODOLOGY

A. System Overview

This section presents an overview of the proposed heading calibration framework, which integrates Wi-Fi fingerprinting-based correction and QSMFI detection. The system is designed

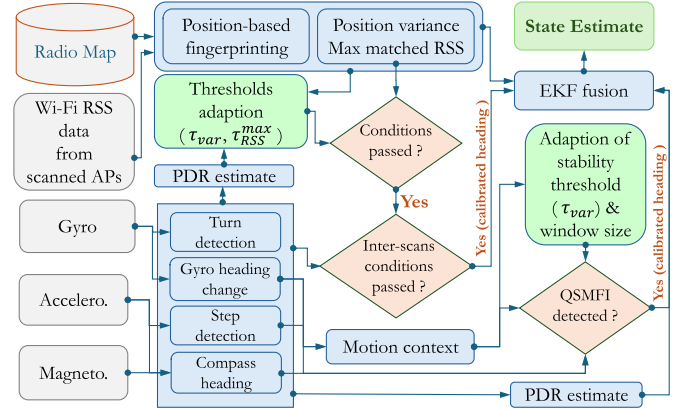


Fig. 1. System framework integrating sensor-based motion analysis, QSMFI detection, and Wi-Fi fingerprinting for heading calibration and EKF-based positioning.

to operate in real time on smartphones using built-in inertial and magnetic sensors along with periodic Wi-Fi scans. As illustrated in Fig. 1, the proposed approach consists of three primary modules: 1) adaptive Wi-Fi fingerprinting quality assessment and heading correction; 2) context-aware QSMFI detection and validation for absolute heading updates; and 3) integration of corrected headings into a PDR system and a sensor fusion framework via an EKF. Each module is elaborated in Sections III-B–III-D, detailing the algorithms, reliability checks, and adaptive mechanisms that ensure robustness and scalability across diverse indoor environments.

B. Heading Calibration Using Accurate Wi-Fi Solutions Triggered While Walking in Straight Portions

To opportunistically calibrate heading during walking segments, we exploit reliable positioning outputs from Wi-Fi fingerprinting, particularly during straight-line motion where heading is most stable. The Wi-Fi fingerprinting methodology employed in this research is based on a weighted k -nearest neighbor (WKNN) algorithm that utilizes observed RSS measurements from nearby Wi-Fi APs. The positioning estimation leverages both the variance of fingerprinting positions and the maximum observed RSS to enhance the reliability and

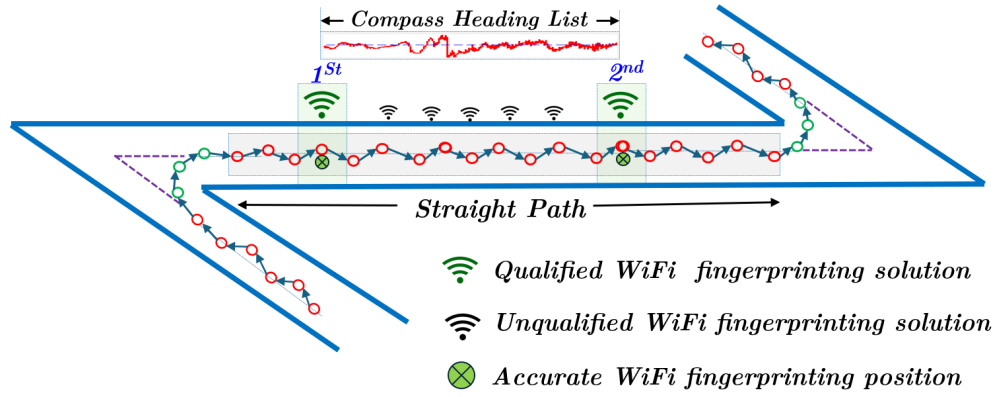


Fig. 2. Conceptual diagram illustrating heading calibration using qualified Wi-Fi fingerprinting positions along a straight path.

accuracy of the solution. The overall approach comprises the following key steps. Fig. 2 presents a conceptual diagram illustrating heading calibration using qualified Wi-Fi fingerprinting positions along a straight path.

1) *RSS Matching and Position Estimation*: Given real-time online RSS measurements $\mathbf{O} = \{o_1, o_2, \dots, o_M\}$ from M APs, the Euclidean distance (D_i) between the online RSS vector and the offline RSS fingerprints stored at each reference point (RP) i is computed as follows:

$$D_i = \sqrt{\sum_{j=1}^M (o_j - r_{ij})^2}, \quad i = 1, 2, \dots, N \quad (1)$$

where r_{ij} is the offline RSS value for AP j at RP i , and N represents the total number of available RPs. After calculating these distances, the closest K RPs with the smallest distances are selected for further processing. The user's estimated fingerprint position $\mathbf{P}_{\text{fps}} = (X_{\text{fps}}, Y_{\text{fps}})$ is computed by calculating the weighted average of the coordinates of the K closest RPs, where each RP i has known coordinates $\mathbf{p}_i = (x_i, y_i)$ and the weight w_i for each RP is inversely proportional to its Euclidean distance D_i

$$\mathbf{P}_{\text{fps}} = \frac{\sum_{i=1}^K w_i \mathbf{p}_i}{\sum_{i=1}^K w_i}, \quad w_i = \begin{cases} \frac{1}{D_i}, & \text{if } D_i \neq 0 \\ 100, & \text{if } D_i = 0. \end{cases} \quad (2)$$

2) *Variance (Spatial Dispersion)-Based Assessment of Fingerprinting Accuracy*: The reliability of the position estimate is quantified by the fingerprinting variance σ_{fps}^2 , which measures the weighted dispersion of the K nearest RPs around the estimated location

$$\sigma_{\text{fps}}^2 = \frac{\sum_{i=1}^K w_i [(X_{\text{fps}} - x_i)^2 + (Y_{\text{fps}} - y_i)^2]}{\sum_{i=1}^K w_i} \quad (3)$$

where σ_{fps}^2 is the weighted spatial variance of the fingerprinting solution, $(X_{\text{fps}}, Y_{\text{fps}})$ is the final fingerprint-based position estimate, (x_i, y_i) is the i th candidate location, w_i is its associated weight, and K is the number of candidates used in the reliability assessment. However, relying exclusively on variance can sometimes introduce ambiguity and lead to faulty reliability

assessments, as a low variance does not necessarily imply high accuracy, due to the following aspects.

- Spatial bias*: Tightly clustered but spatially biased neighbors can yield confident yet incorrect estimates.
- Geometric ambiguity*: Symmetric environments may cause low-variance matches with incorrect locations.
- Uneven AP distribution*: Local clustering in sparse regions can falsely suggest reliability.

3) *Mitigating Variance Ambiguity and Faulty Reliability Assessments Through Signal-Aided Confidence*: To avoid faulty reliability assessments caused by relying solely on variance, and to mitigate its inherent limitations, we propose augmenting the localization reliability estimation by incorporating the maximum online RSS observed among the matched APs (RSS_{max}) as a complementary criterion as follows:

$$\text{RSS}_{\text{max}} = \max_{j \in \text{matched APs}} (o_j) \quad (4)$$

where RSS_{max} is the maximum RSS among the matched APs and o_j is the observed RSS of the j th matched AP. The incorporation of RSS_{max} as a complementary reliability indicator in Wi-Fi fingerprinting is grounded in several key considerations. First, low values of RSS_{max} imply that all received signals are weak, which typically indicates poor signal quality and increases the risk of generating ambiguous fingerprint matches. In such cases, the system may struggle to differentiate between candidate locations due to the lack of dominant signal features. Conversely, the presence of at least one strong RSS measurement, even when other signals are weak, can provide decisive discriminative power, enhancing the distinctiveness and reliability of the fingerprint match. Second, higher RSS values generally suggest physical proximity to one or more APs, which tends to improve measurement stability and consistency across both offline and online phases. This proximity is also often associated with line-of-sight (LOS) conditions, further contributing to the trustworthiness of the observation. Finally, in contrast to commonly used variance-based metrics, which quantify spatial dispersion across neighboring fingerprints, RSS_{max} emphasizes the signal-centric aspect of confidence. This makes it particularly valuable for identifying reliable matches, as it can effectively distinguish between strong, credible fingerprints and those dominated by weak or noisy

measurements. Consequently, when both a low σ_{fps}^2 and a strong RSS_{max} are observed simultaneously, the confidence of the fingerprinting accuracy should be high.

4) *Automatic Estimation and Adaptation of Thresholds for Qualifying Wi-Fi Fingerprinting Solutions for Heading Calibration:* To determine whether a fingerprinting solution is sufficiently accurate to serve as a reference for heading calibration, it is essential to define thresholds for the two key reliability indicators: the σ_{fps}^2 and RSS_{max} . Instead of relying on manually tuned parameters, we propose an automatic, environment-adaptive framework that estimates these thresholds dynamically based on early observations. This design enables the method to operate robustly across diverse indoor scenarios without prior calibration or expert intervention.

During the initialization phase, the system collects a set of early fingerprinting-based estimates to characterize the empirical behavior of the two key reliability indicators. Let $\{(\sigma_{\text{fps}}^2)^{(1)}, \dots, (\sigma_{\text{fps}}^2)^{(N)}\}$ and $\{\text{RSS}_{\text{max}}^{(1)}, \dots, \text{RSS}_{\text{max}}^{(N)}\}$ denote the initial N observed values of fingerprinting variance and the corresponding maximum matched RSSs respectively. Based on this initial set, the method derives the empirical distributions of both parameters and defines quantitative ranges using fixed quantiles. Concretely, the “low-variance” group is formed by taking approximately the lowest 30% of the observed variance values, while the “high-RSS” group is formed by taking approximately the highest 30% of the observed maximum RSS values. The variance threshold $\tau_{\text{var}}^{\text{wifi}}$ is then obtained by averaging the variances in the low-variance group and adding a small margin δ_{σ} to form a conservative upper bound

$$\tau_{\text{var}}^{\text{wifi}} = \frac{1}{|\mathcal{L}_{\sigma}|} \sum_{i \in \mathcal{L}_{\sigma}} (\sigma_{\text{fps}}^2)^{(i)} + \delta_{\sigma} \quad (5)$$

where $\mathcal{L}_{\sigma}$ denotes the index set corresponding to samples in the lowest-variance range. Similarly, the threshold for signal strength is determined by computing the average RSS_{max} within the strongest observed range, then subtracting a small margin δ_{rss} to form a conservative lower bound

$$\tau_{\text{rss}}^{\text{max}} = \frac{1}{|\mathcal{H}_{\text{rss}}|} \sum_{i \in \mathcal{H}_{\text{rss}}} \text{RSS}_{\text{max}}^{(i)} - \delta_{\text{rss}} \quad (6)$$

where $\mathcal{H}_{\text{rss}}$ denotes the index set corresponding to samples in the strongest (i.e., highest) RSS_{max} range. A fingerprinting solution is considered qualified for heading calibration if it satisfies both of the following conditions:

$$\sigma_{\text{fps}}^2 \leq \tau_{\text{var}}^{\text{wifi}} \quad \text{and} \quad \text{RSS}_{\text{max}} \geq \tau_{\text{rss}}^{\text{max}}. \quad (7)$$

In this formulation, δ_{σ} and δ_{rss} are not additional measurement errors, but conservative safety margins that stabilize the automatically estimated thresholds. The initial N observations provide only a finite sample of the environment, and Wi-Fi signals exhibit temporal variability due to dynamic factors such as door movements, pedestrian traffic, and changes in device orientation. If the margins were set to zero, $\tau_{\text{var}}^{\text{wifi}}$ and $\tau_{\text{rss}}^{\text{max}}$ would coincide with the empirical means of their “good” ranges, which is optimistic and tends to accept borderline solutions that are only momentarily reliable. The positive margin δ_{σ} shifts the variance threshold to favor the most stable low-variance cases, while the subtraction margin δ_{rss}

Algorithm 1 Wi-Fi Fingerprinting Qualification Check With Adaptive Thresholds

Input: $\sigma_{\text{fps}}^2, \text{RSS}_{\text{max}}, \bar{\Delta}s_{\text{cal}}, \bar{d}_{\text{diff}}$
Parameters: $\tau_{\text{var}}^{\text{wifi}}, \tau_{\text{rss}}^{\text{max}}$ (adaptive)
Output: isQualifiedSolution (Boolean)

```

1: isQualifiedSolution  $\leftarrow$  false
2: if  $\text{RSS}_{\text{max}} > \tau_{\text{rss}}^{\text{max}}$  and  $\sigma_{\text{fps}}^2 < \tau_{\text{var}}^{\text{wifi}}$  then
3:   | isQualifiedSolution  $\leftarrow$  true
4: else
5:   |  $\triangleright$  Adaptive threshold tuning
6:   | Compute  $\bar{\Delta}s_{\text{cal}}$  and  $\bar{d}_{\text{diff}}$   $\triangleright$  Eqs. 8, 9
7:   | if  $f_{\text{cal}}$  is low or  $\bar{\Delta}s_{\text{cal}}$  is high or  $\bar{d}_{\text{diff}}$  is large then
8:   |   |  $\delta_{\sigma} \leftarrow \delta_{\sigma} + \epsilon_{\sigma}$ 
9:   |   |  $\delta_{\text{rss}} \leftarrow \delta_{\text{rss}} - \epsilon_{\text{rss}}$ 
10:  |   | Update  $\tau_{\text{var}}^{\text{wifi}}$  and  $\tau_{\text{rss}}^{\text{max}}$   $\triangleright$  Eqs. 5, 6
11: return isQualifiedSolution

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prevents $\tau_{\text{rss}}^{\text{max}}$ from being inflated by a few abnormally strong measurements. As a result, only the most reliable subset of fingerprinting solutions passes the gate in (7), reducing the false acceptance of unreliable calibrations. The margins are further adapted online in Algorithm 1, based on the observed calibration spacing and the disagreement between PDR and fingerprinting positions.

Once initialized, the framework enters a continuous evaluation phase in which it assesses the quality of the selected thresholds. Instead of relying on a fixed time window, the evaluation is performed over a moving window of walking steps, denoted by S steps. This step-based windowing reflects the natural granularity of user motion in indoor positioning scenarios. The step index is derived from the same accelerometer-based step detector used in the PDR module. Specifically, the PDR module detects steps using a standard peak detector applied to the smoothed acceleration in the vertical (z) direction relative to the Earth reference frame, after low-pass filtering and enforcing a refractory interval based on the gait cycle. This approach provides reliable step counts for typical human walking and is widely adopted in smartphone-based PDR.

First, the method evaluates the spatial spacing between successive calibrations by computing the average number of walking steps separating consecutive calibration events. This spacing metric reflects how densely or sparsely the qualified calibrations are distributed along the user’s trajectory

$$\bar{\Delta}s_{\text{cal}} = \frac{1}{N_{\text{cal}} - 1} \sum_{j=1}^{N_{\text{cal}}-1} (s_{j+1} - s_j) \quad (8)$$

where s_j denotes the cumulative step index at which the j th successful heading calibration occurred. A consistently low $\bar{\Delta}s_{\text{cal}}$ indicates frequent and well-distributed calibration opportunities, while a high value suggests sparse or infrequent calibration support. Second, it evaluates the average positional disagreement between the PDR-fused trajectory using the calibrated heading $\mathbf{p}_{\text{PDR}}$ and the fingerprinting-based position

$\mathbf{p}_{\text{FPS}}$ at the moments of calibration

$$\bar{d}_{\text{diff}} = \frac{1}{N_{\text{cal}}} \sum_{j=1}^{N_{\text{cal}}} \left\| \mathbf{p}_{\text{PDR}}^{(j)} - \mathbf{p}_{\text{FPS}}^{(j)} \right\|. \quad (9)$$

The step-based windowing and the spacing metric in (8) therefore use the cumulative step counts only as a relative measure of how densely the calibration events are distributed along the trajectory. Moderate step-detection errors mainly introduce a small scaling in $\bar{\Delta}s_{\text{cal}}$ and a slight change in the effective window length S , while the key performance indicator $\bar{d}_{\text{diff}}$ in (9) is expressed in meters and remains independent of the step count. As a result, the adaptive update of the thresholds is driven predominantly by the geometric disagreement between PDR and fingerprinting, and is robust to typical levels of step-detection error in smartphone-based PDR.

If the calibration frequency is sufficiently high, the spacing between successive calibrations remains consistent (i.e., $\bar{\Delta}s_{\text{cal}}$ remains low), and the positional disagreement stays within acceptable bounds, the thresholds are considered reliable and retained. Conversely, if calibrations become sparse, reflected by a high $\bar{\Delta}s_{\text{cal}}$, or if the disagreement between the PDR and fingerprinting positions increases, the method adapts the thresholds by adjusting the margins δ_{σ} and δ_{rss} based on recent step-wise observations. This adaptive mechanism ensures that the qualification of fingerprinting solutions remains accurate and resilient under dynamically changing indoor environments. Based on the estimated thresholds, a fingerprinting solution is considered suitable for heading calibration only if it passes the quality gate defined in Algorithm 1, which jointly evaluates the observed signal strength and spatial stability of the estimated position.

Algorithm 2 Wi-Fi-Based Heading Calibration in Straight Segments

Input: Wi-Fi solution F_i , step s_i , turn t_i
compass heading h_i^{mag}

Output: $h_{\text{calibrated}}$

```

1: Initialize flags and  $h_{\text{calibrated}} \leftarrow \text{null}$ 
2: while magneticFieldSensorChanged do
3:   Check if  $F_i$  is new and qualified ▷ Alg. 1
4:   if First qualified solution not yet detected then
5:     if Current solution is qualified then
6:       Record  $P_1: (s_1, t_1) \leftarrow (s_i, t_i)$ 
7:       initialize  $\mathbf{v}^{\text{compass}}$ 
8:   else
9:     Collect compass headings in  $\mathbf{v}^{\text{compass}}$ 
10:    if Second qualified solution detected then
11:      Record  $P_2: (s_2, t_2) \leftarrow (s_i, t_i)$ 
12:       $h_{\text{calibrated}} \leftarrow \text{CalibrationConditions}()$  ▷ Alg. 3
13: return  $h_{\text{calibrated}}$ 

```

5) *Multilayer Validation via Step Distance and Compass Heading:* Upon obtaining the first qualified solution (P_1), the method records the corresponding pedestrian step and turn numbers (s_1 and t_1). It simultaneously initializes a compass heading buffer ($\mathbf{v}^{\text{compass}}$) to accumulate compass readings collected until a second qualified Wi-Fi solution (P_2) is detected.

Algorithm 3 Multilayer Calibration Conditions

Input: $P_1, P_2, s_1, s_2, t_1, t_2, \mathbf{v}^{\text{compass}}, h_i^{\text{mag}}$

Parameters: $\tau_{\delta_s} = 20\% \cdot \delta_d, \tau_s = 3, \tau_{\delta_h} = 10^\circ$

Output: $h_{\text{calibrated}}$

```

1: Procedures:
2: if isSecondQualifiedSolDetected then
3:    $h_{\text{calibrated}} \leftarrow \text{null}$ 
4:    $\Delta t = t_2 - t_1$  ▷ Turns between solutions
5:   if  $\Delta t < 1$  then ▷ Condition 1: No turns
6:      $\Delta s = s_2 - s_1$  ▷ Steps between solutions
7:      $\Delta x = x_{P_2} - x_{P_1}, \Delta y = y_{P_2} - y_{P_1}$ 
8:      $d_{1-2} = \sqrt{(\Delta x)^2 + (\Delta y)^2}$  ▷ Distance from Wi-Fi
9:      $ds_{1-2} = \Delta s \cdot \bar{l}_s$  ▷ Distance from steps
10:     $\delta_d = |d_{1-2} - ds_{1-2}|$ 
11:    if  $\Delta s > \tau_s \vee \delta_d \leq \tau_{\delta_s}$  then
12:      ▷ Condition 2: Sufficient displacement
13:       $h_{1-2} = (\text{atan2}(\Delta x, \Delta y) \cdot 180/\pi + 360) \bmod 360$ 
14:       $\delta_h^{\text{mo}} = |\text{mode}(\mathbf{v}^{\text{compass}}) - h_{1-2}|$ 
15:       $\delta_h^{\text{m}} = |\text{mean}(\mathbf{v}^{\text{compass}}) - h_{1-2}|$ 
16:       $\delta_h^{\text{i}} = |h_i^{\text{mag}} - h_{1-2}|$ 
17:      if  $(\delta_h^{\text{m}} < \tau_{\delta_h} \vee \delta_h^{\text{mo}} < \tau_{\delta_h}) \wedge (\delta_h^{\text{i}} < \tau_{\delta_h})$  then
18:        ▷ Condition 3: Heading consistency
19:         $h_{\text{calibrated}} \leftarrow h_{1-2}$  ▷ All conditions met
20:         $P_1 \leftarrow P_2, s_1 \leftarrow s_2, t_1 \leftarrow t_2$ 
21:         $\mathbf{v}^{\text{compass}} \leftarrow \emptyset$ 
22:        isFirstQualifiedSolDetected  $\leftarrow \text{true}$ ,
23:        isSecondQualifiedSolDetected  $\leftarrow \text{false}$ 
24:    else
25:      isSecondQualifiedSolDetected  $\leftarrow \text{false}$ 
26:    else ▷ Turn detected: reset
27:       $P_1 \leftarrow P_2, s_1 \leftarrow s_2, t_1 \leftarrow t_2, \mathbf{v}^{\text{compass}} \leftarrow \emptyset$ 
28:      isFirstQualifiedSolDetected  $\leftarrow \text{true}$ ,
29:      isSecondQualifiedSolDetected  $\leftarrow \text{false}$ ,
30: return  $h_{\text{calibrated}}$ 

```

At this stage, a multilayer check (see Algorithm 3) is executed to assess the robustness of the calibration opportunity. This check enforces three key conditions: 1) fewer than one turn ($\Delta t < 1$) must occur between the two qualified Wi-Fi solutions, indicating a straight walking segment; 2) the step-based displacement must closely match the Wi-Fi-derived displacement (δ_d), within a tolerance τ_{δ_s} , or the step count must exceed a minimum threshold ($\tau_s = 3$); and 3) compass-based heading estimates (mean, mode, and instantaneous) must all deviate by less than $\tau_{\delta_h} = 10^\circ$ from the position-derived heading between P_1 and P_2 .

If all these conditions are met, a calibrated heading ($h_{\text{calibrated}}$) is computed using the positions derived from two accurate Wi-Fi fingerprinting solutions. This calibrated heading is considered reliable and subsequently used to update the pedestrian's orientation.

C. Enhanced Heading Calibration Based on QSMFI

QSMFIs are detected using a sliding window over recent compass heading samples (in degrees). Let H_i denote the headings within a window W of size $N = |W|$. The window is

classified as quasi-static if the heading variance falls below a threshold $\tau_{\text{var}}^{\text{mag}}$

$$\text{Var}(H) = \frac{1}{|W|} \sum_{i \in W} (H_i - \bar{H})^2 < \tau_{\text{var}}^{\text{mag}} \quad (10)$$

where $\bar{H}$ is the mean heading in the window. Low variance indicates a stable compass direction and a potential QSMFI for heading correction.

Magnetometer readings are preprocessed to reduce bias before entering this pipeline. We use the calibrated outputs provided by the Android sensor framework (factory hard- and soft-iron compensation) and perform a short 3-D “figure-eight” motion before each recording session to refresh the device-level calibration. The calibrated magnetic field is then tilt-compensated using the current attitude from accelerometer–gyroscope fusion, and only the horizontal components are used to derive the compass heading H_i .

Both the window size and the variance threshold are adapted to the motion context. During stationary periods, longer windows and stricter thresholds are used to emphasize stability, whereas during walking, shorter windows and slightly relaxed thresholds capture brief steady intervals without over-filtering locomotion-induced fluctuations. In addition, each candidate QSMFI is validated through a gyroscope-compass consistency check: if the difference between magnetometer heading and integrated gyroscope heading exceeds a predefined limit, the interval is discarded. This combination of bias-aware preprocessing, adaptive parameterization, and cross-sensor validation yields reliable QSMFIs for heading calibration and improves robustness to magnetic disturbances.

1) *Stationary and Walking Mode Differentiation*: The algorithm continuously classifies the user state as stationary or walking to apply the appropriate sliding window and variance strategy. Stationary mode is declared when no new steps are detected and the orientation change remains negligible, i.e., the gyroscope reports (near-)zero angular velocity over a minimum duration in a sliding fashion. When significant motion is detected (steps from the accelerometer or $\Delta\Theta_{\text{gyro}}$ exceeding τ_{gyro}), the system switches to walking mode, and the algorithm applies context-specific window sizes and variance limits for heading calibration.

2) *Sliding-Window Sizing Strategies*: Sliding-window sizes are adapted differently in stationary and walking modes to enable reliable QSMFI detection.

In stationary mode with negligible rotation, a larger sliding-window W_s collects magnetometer-based compass headings $H_{\text{comp}}(t)$. The cumulative yaw rotation over the window is evaluated from the gyroscope

$$\Delta\Theta_{\text{gyro}}(W_s) = \left| \int_{t_0}^{t_0+T_s} \omega_z(\tau) d\tau \right| \quad (11)$$

where $\omega_z(\tau)$ is the yaw-rate, t_0 is the window start, and T_s is the duration. The window W_s is incrementally enlarged as long as

$$\begin{aligned} \Delta\Theta_{\text{gyro}}(W_s) &< \tau_{\text{gyro}} \\ \text{Var}(H_i) &< \tau_{\text{var}}^{\text{mag}} \quad \forall i \in W_s \end{aligned} \quad (12)$$

hold, where τ_{gyro} is a small angular limit and $\tau_{\text{var}}^{\text{mag}}$ the adaptive variance threshold from (10). Once both conditions are met, the interval is accepted as a QSMFI and the stable heading $\hat{\Psi}_s$ is obtained via circular averaging

$$\hat{\Psi}_s = \text{atan2} \left(\frac{1}{N_s} \sum_{i \in W_s} \sin H_i, \frac{1}{N_s} \sum_{i \in W_s} \cos H_i \right) \quad (13)$$

with N_s samples in W_s . The resulting $\hat{\Psi}_s$ is used for recalibration if the window also passes the faulty-period check in Section III-C4; otherwise, the window is reset.

Algorithm 4 QSMFI-Based Heading Calibration With Adaptive Thresholds

Input: $H_{\text{comp}}(t)$, $\omega_z(t)$
Output: $\hat{\Psi}_{\text{ref}}$

```

1: Initialize:  $M \leftarrow \emptyset$ ,  $\hat{\Psi}_{\text{ref}} \leftarrow \emptyset$ ,  $W_s, W_w \leftarrow \emptyset$ 
2: Set thresholds:  $\tau_{\text{gyro}}$ ,  $\tau_{\text{var}}^{\text{mag}}$ ,  $\tau_{\Delta}$ , QSMFI history
3: while new sensor data at  $t_k$  do
4:   Read  $H_{\text{comp}}(t_k)$ ,  $\omega_z(t_k)$ 
5:   ▷ Mode detection
6:   if  $\Delta\Theta_{\text{gyro}}(t_k) < \tau_{\text{static}}$  and  $\Delta\text{Step}(t_k) = 0$  then
7:      $M \leftarrow \text{Static}$ 
8:   else
9:      $M \leftarrow \text{Walking}$ 
10:  if  $M = \text{Static}$  then
11:    Append  $H_{\text{comp}}(t_k)$  to  $W_s$ 
12:    if  $\Delta\Theta_{\text{gyro}}(W_s) < \tau_{\text{gyro}}$  and  $\text{Var}(W_s) < \tau_{\text{var}}^{\text{mag}}$  then
13:      Expand  $W_s$ , compute  $\hat{\Psi}_s$  (Eq. 13)
14:      NewQSMFI  $\leftarrow (\hat{\Psi}_s, t_k)$ , reset  $W_s$ 
15:    else if  $\Delta\Theta_{\text{gyro}}(W_s) \geq \tau_{\text{static}}$  then
16:      Reset  $W_s$ 
17:  else ▷ Walking mode
18:    Slide and append  $H_{\text{comp}}(t_k)$  to  $W_w$ 
19:    if  $\text{Var}(W_w) < \tau_{\text{var}}^{\text{mag}}$  then
20:      Mark  $W_w$  as stable, extend if needed
21:    else if last  $W_w$  was stable then
22:      Extract  $W_w^*$ , compute  $\hat{\Psi}_w$ 
23:      NewQSMFI  $\leftarrow (\hat{\Psi}_w, t_k)$ , reset  $W_w$ 
24:    Adapt window size (Eq. 14)
25:    Adapt variance threshold (Eq. 18)
26:    ▷ Gyro-compass cross-validation
27:  if NewQSMFI exists then
28:    if  $\hat{\Psi}_{\text{ref}} = \text{None}$  then
29:      Accept:  $\hat{\Psi}_{\text{ref}} \leftarrow \hat{\Psi}_{\text{new}}$ 
30:    else
31:       $\Delta\Psi_{\text{mag}} = \text{wrap}(\hat{\Psi}_{\text{new}} - \hat{\Psi}_{\text{ref}})$ 
32:       $\Delta\Psi_{\text{gyro}} = \int_{t_{\text{ref}}}^{t_{\text{new}}} \omega_z(t) dt$ 
33:      if  $|\Delta\Psi_{\text{mag}} - \Delta\Psi_{\text{gyro}}| < \tau_{\Delta}$  then
34:        Accept: update  $\hat{\Psi}_{\text{ref}}$ , history (Eq. 18)
35:      else
36:        Reject: apply penalty (Eq. 18)

```

During walking, QSMFIs may still occur when the user moves through magnetically stable regions. An adaptive window W_w of size N_w is used, initialized from empirical analysis. If the heading variance repeatedly exceeds $\tau_{\text{var}}^{\text{mag}}$ but the change

between successive QSMFI candidates $\hat{\Psi}_k$ and $\hat{\Psi}_{k-1}$ remains consistent with the integrated gyro heading $\Delta\Psi_{\text{gyro}}$, the window size is increased

$$\text{Var}(H_i) > \tau_{\text{var}}^{\text{mag}} \quad \text{and} \quad |\text{wrap}(\hat{\Psi}_k - \hat{\Psi}_{k-1}) - \Delta\Psi_{\text{gyro}}| < \tau_{\Delta} \quad (14)$$

$$N_w \leftarrow N_w + \delta_N \quad (15)$$

where $\Delta\Psi_{\text{gyro}} = \int_{t_{k-1}}^{t_k} \omega_z(t) dt$, $\text{wrap}(\cdot)$ bounds angles to $[-\pi, \pi]$, and τ_{Δ} is a small agreement threshold. If no valid QSMFI is found after reaching N_{max} , the variance limit is slightly relaxed

$$\tau_{\text{var}}^{\text{mag}} \leftarrow \tau_{\text{var}}^{\text{mag}} + \delta_{\tau} \quad (16)$$

allowing mild instability. This dual adaptation keeps the detector both responsive and robust across varying motion and magnetic conditions.

3) *Adaptive Variance Threshold for Robust QSMFI Detection*: To maintain robust QSMFI detection under varying conditions, the heading variance threshold $\tau_{\text{var}}^{\text{mag}}$ is adapted using the recent success rate of QSMFI candidates. Let r_t denote the average acceptance rate over the past M intervals

$$r_t = \frac{1}{M} \sum_{m=1}^M \mathbb{I}_{\text{accept}}^{(t-m)} \quad (17)$$

where $\mathbb{I}_{\text{accept}}^{(t-m)} = 1$ if the QSMFI at time $t - m$ passed gyro–compass validation and 0 otherwise. The variance threshold is then updated as

$$\tau_{\text{var}}^{\text{mag},(t+1)} = \tau_{\text{var}}^{\text{mag},(t)} + \varepsilon (\delta_{\text{inc}} (1 - r_t) - \delta_{\text{dec}} r_t) \quad (18)$$

with positive constants δ_{inc} , δ_{dec} and a small learning rate ε . Frequent acceptance (high r_t) makes the detector stricter by decreasing $\tau_{\text{var}}^{\text{mag},(t)}$, while frequent rejection (low r_t) relaxes it. The updated threshold is clipped to $[\tau_{\text{min}}, \tau_{\text{max}}]$ and uses larger base values in walking mode than in stationary mode. This simple feedback rule steers $\tau_{\text{var}}^{\text{mag},(t)}$ toward a balance between sensitivity and reliability.

4) *Rejection of Faulty QSMFI via Gyroscope–Magnetometer Consistency Check*: Low variance does not always imply a reliable magnetic reference, especially in distorted magnetic environments. Therefore, each QSMFI candidate is further checked against the gyroscope-integrated heading. Instead of relying on a single interval, the check aggregates S successive QSMFIs. The cumulative magnetic heading change is

$$\Delta\Psi_{\text{mag}}^{\text{total}} = \text{wrap} \left(\sum_{i=1}^S (\hat{\Psi}_{k-S+i} - \hat{\Psi}_{k-S+i-1}) \right) \quad (19)$$

which is compared with the integrated gyro heading change $\Delta\Psi_{\text{gyro}}$ over $[t_{k-S}, t_k]$. The QSMFI is accepted only if

$$|\Delta\Psi_{\text{mag}}^{\text{total}} - \Delta\Psi_{\text{gyro}}| < \tau_{\Delta}. \quad (20)$$

A larger discrepancy indicates magnetic interference or violation of the quasi-static assumption, and the affected intervals are rejected or skipped for recalibration. In walking mode, the increment δ_{τ} in (16) is set to a small fraction of the nominal variance range and is applied only when no QSMFI is found after the window size reaches N_{max} . The updated

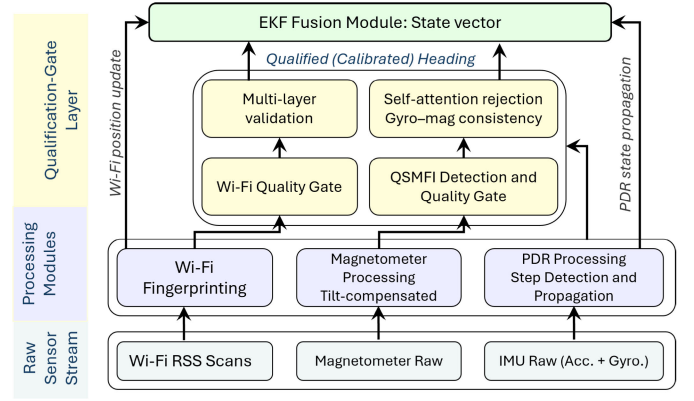


Fig. 3. Fusion flow for the proposed EKF-based positioning method.

threshold $\tau_{\text{var}}^{\text{mag}}$ is clipped to $[\tau_{\text{min}}, \tau_{\text{max}}]$, and all candidate intervals must still satisfy the gyro–magnetometer consistency condition in (20). Together with the history-based update in (18), this ensures that the occasional relaxation introduced by δ_{τ} improves sensitivity in challenging environments without significantly increasing the risk of false QSMFI detections

D. Integrated Positioning Based on EKF

To counteract the cumulative heading and positional drift of PDR over time, and to exploit PDR reliably during outages of the fingerprinting-based Wi-Fi RSS solution, we employ a loosely coupled fusion strategy based on an EKF, as shown in Fig. 3. In this scheme, PDR provides continuous state propagation, while high-confidence Wi-Fi fingerprinting solutions and robustly validated QSMFIs are injected as intermittent pseudo-measurements when they pass the reliability gate. This gate is designed to prevent unreliable updates under irregular handheld motion by rejecting calibration opportunities that violate straight-motion and displacement-consistency criteria. The filter maintains an internal state vector $\mathbf{x}_t = [x_t, y_t, s_t, h_t]^T$, representing the pedestrian's 2-D coordinates (x_t, y_t) , stride-length correction factor s_t , and instantaneous heading angle h_t . This state is updated iteratively through a recursive prediction–correction cycle: inertial displacement updates are applied at each detected step, and whenever a qualified Wi-Fi/QSMFI-based calibration becomes available, it is used to correct the heading (and indirectly the position) inside the EKF. Step detection is based on peak–valley analysis of the Earth-frame vertical acceleration. Raw accelerometer readings in the device coordinate system, a_t^{dcs} , are first low-pass filtered to estimate gravity

$$g_t^{\text{dcs}} = \alpha g_{t-1}^{\text{dcs}} - (1 - \alpha) a_t^{\text{dcs}} \quad (21)$$

where g_t^{dcs} is the gravitational component and α is a filter constant (e.g., $\alpha = 0.8$ at 50 Hz). Linear acceleration in the navigation frame is then obtained via the attitude-dependent rotation matrix C_t

$$a_t^{\text{ncs}} = C_t (a_t^{\text{dcs}} - g_t^{\text{dcs}}) \quad (22)$$

and its vertical component is used to locate peaks and valleys. A sample pair (valley, peak) is accepted as a valid step i only if the amplitude and duration satisfy

$$\delta a_i = a_{z,\text{peak},i}^{\text{ncs}} - a_{z,\text{valley},i}^{\text{ncs}} \geq T_a \quad (23)$$

$$\Delta t_i = t_{\text{peak},i} - t_{\text{peak},i-1} \in [\Delta t_{\min}, \Delta t_{\max}] \quad (24)$$

where T_a is an acceleration threshold (about 0.65 m/s^2) and $[\Delta t_{\min}, \Delta t_{\max}]$ bounds the plausible step period (about 100 ms to 1.6 s). Regarding the robustness of the step detection under unconstrained smartphone usage, although peak-based step detectors can be sensitive when applied to raw device-axis acceleration, our implementation detects steps using the Earth-frame vertical component after gravity removal and attitude-based rotation [see (21) and (22)]. This design reduces dependence on the instantaneous phone pose across common carrying modes. To mitigate false triggers caused by nongait hand motions (e.g., brief shaking or phone-to-ear transitions), a candidate step is accepted only when it satisfies joint amplitude and cadence constraints [see (23) and (24)] and preserves the expected valley–peak alternation, while short-interval jitter is suppressed using a short temporal window. More details and step-detection evaluation results can be found in [32]. Notably, the step-detection module is replaceable, and more recent carrying-mode-aware or learning-based detectors (see [33], [34]) can be integrated to further strengthen robustness, without changing the proposed heading calibration gate or the EKF fusion structure. For each accepted step, the step length r_i is estimated using an empirical fourth-root model

$$r_i = m \delta a_i^{1/4} \quad (25)$$

where m is a calibration coefficient and $0.6 \leq r_i \leq 0.9$ enforces a plausible range of step lengths. At each detected step, the EKF executes a prediction based on these inertial inputs.

When an accurate external heading calibration from Wi-Fi fingerprinting (see Section III-B) or a verified QSMFI-based heading correction (see Section III-C) is available, the heading component of the state vector is directly updated through a small increment δ_h

$$h_t = h_{t-1} + \delta_h \quad (26)$$

where δ_h denotes the calibrated offset. To reflect high confidence in these corrections, the EKF measurement noise for heading is set to a small value (e.g., $r_h = 0.001$).

The corrected heading is then used in the position prediction. Considering stride-length corrections and heading refinements, the EKF predicts the next state as

$$\begin{aligned} x_t &= x_{t-1} + (1 + s_{t-1}) d \sin(h_t) + \eta_x \\ y_t &= y_{t-1} + (1 + s_{t-1}) d \cos(h_t) + \eta_y \\ s_t &= s_{t-1} + \eta_s \end{aligned} \quad (27)$$

where d is the nominal step length, and η_x , η_y , and η_s are Gaussian process noise terms capturing sensor errors and gait variability. To further mitigate positional drift, absolute positions from validated Wi-Fi fingerprinting solutions are fused when available. If a fingerprinting-based position $\mathbf{z}_t =$

$[x_t^{\text{obs}}, y_t^{\text{obs}}]^T$ meets the reliability criteria in Section III-B, it is incorporated via

$$\mathbf{z}_t = \mathbf{H}\mathbf{x}_t + \mathbf{v}_t, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (28)$$

where $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ and $\mathbf{R}$ reflects the fingerprinting position uncertainty. The correction step is then performed in the standard way

$$\begin{aligned} \mathbf{K}_t &= \mathbf{P}_{t|t-1} \mathbf{H}^T (\mathbf{H} \mathbf{P}_{t|t-1} \mathbf{H}^T + \mathbf{R})^{-1} \\ \hat{\mathbf{x}}_t &= \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t (\mathbf{z}_t - \mathbf{H} \hat{\mathbf{x}}_{t|t-1}) \\ \mathbf{P}_t &= (\mathbf{I} - \mathbf{K}_t \mathbf{H}) \mathbf{P}_{t|t-1} \end{aligned} \quad (29)$$

where $\hat{\mathbf{x}}_t$ is the updated state estimate, $\mathbf{P}_t$ is the associated covariance, and $\mathbf{I}$ is the identity matrix.

IV. EXPERIMENTS, RESULTS, AND EVALUATIONS

A. Experimental Settings

To evaluate the proposed heading calibration framework, we first constructed an offline radio map for the Block Z floors of The Hong Kong Polytechnic University by performing dynamic walking surveys between designated landmarks. We developed a custom Android application in Java that implements the complete system and runs entirely on the smartphone. The app integrates the Google Maps API for trajectory and real-time heading visualization and uses only on-board sensors: magnetometer and accelerometer at 50 Hz, and gyroscope at 100 Hz for accurate heading integration. Wi-Fi scans were automatically triggered every 2–4 s on Huawei Mate 20 Pro devices; on devices with stricter Android scan throttling, the interval could reach 10–20 s. This setup enabled continuous testing in realistic conditions using consumer-grade hardware only.

B. Ground Truth (GT) Establishment

Accurate GT was established using a two-step georeferencing procedure. First, external benchmarks were surveyed in an open area with a high-precision Trimble R10 GNSS receiver to obtain precise geographic coordinates. Second, indoor RPs were measured relative to these benchmarks and integrated into a georeferenced AutoCAD floor plan, which was then overlaid on Google Maps within the custom Android application for precise position tracking.

In the Android app UI, two types of map markers were implemented via the `onMapReady()` method for GT collection. A long press on the georeferenced map inserts an initial marker, and the `onMarkerDragEnd()` callback allows the user to refine its position by dragging it to the exact location on the floor plan. Upon release, the marker is stored as the confirmed GT position. To ensure data integrity, if multiple `onMarkerDragEnd()` events occur while the user remains stationary (no change in step number), only the final marker placement is retained and earlier drag events are discarded.

A dedicated data-writing function records the finalized marker information, including timestamp, coordinates, and associated step number. The tester interactively places or adjusts GT markers at key RPs such as corners, corridor ends,

TABLE II
SAMPLE OF MANUALLY LABELED GT LOCATIONS AND HEADING

GT Point	Timestamp (ms)	Type	Step No.	GT Creation Method	East (m)	North (m)	GT Heading (°)
GT(1)	1746240567147	Start Point	0.5	Location of Marker Created By Long Click [*]	836559.31	818602.90	–
GT(2)	1746240667365	Turn	74.5	Location at End Marker Dragging [†]	836510.50	818617.26	292.21
GT(3)	1746240715817	Turn	117.0	Location at End Marker Dragging [†]	836502.38	818589.13	199.25
...	...	...	...	...	...	...	...
GT(9)	1746240954107	End Point	227.0	Location at End Marker Dragging [†]	836510.01	818616.85	23.19

^{*} This point was manually labeled by long pressing the map to mark the initial position after distinguishing the actual location of the occupied point on the floor plan. [†] These points were manually labeled by dragging the on-screen marker to the end of each observed walking segment. The location of the marker after finishing dragging to the exact location of the occupied point was recorded as the GT location of this turn.

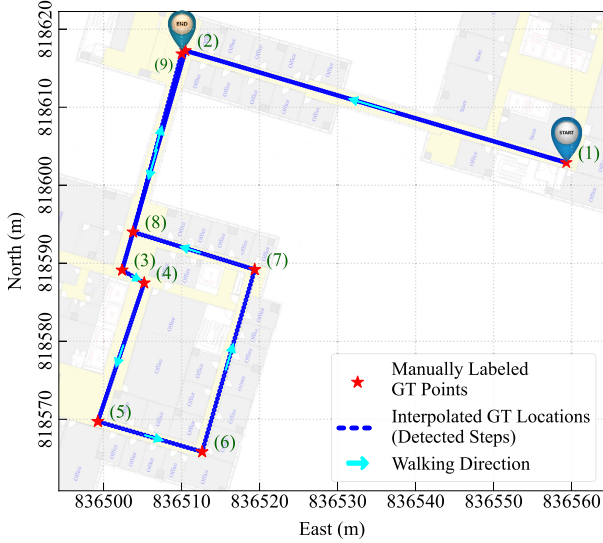


Fig. 4. Reference trajectory used for evaluating both localization and heading estimation accuracy.

and major changes in walking direction. Table II presents an excerpt of the recorded GT data with headings. Intermediate positions between successive GT points are interpolated from the detected number of steps and the known distance between markers, as illustrated in Fig. 4.

C. Test Descriptions and Setup

The experiments were organized around three main objectives as follows.

- 1) *Wi-Fi-Based Heading Calibration*: We evaluated the Wi-Fi-based heading calibration in Section III-B by varying the dual thresholds on fingerprinting variance (σ_{fpi}^2) and maximum RSS (RSS_{max}). For each setting, calibration frequency and heading error were computed with respect to GT.
- 2) *QSMFI-Based Heading Calibration*: We assessed heading correction during QSMFIs under different stability conditions by varying the sliding-window size and variance threshold ($\tau_{\text{var}}^{\text{mag}}$). The role of the adaptive mechanism and the self-attention-based consistency check was analyzed by comparing calibrated headings to GT.
- 3) *Integrated EKF Positioning*: We evaluated the full system, where Wi-Fi- and QSMFI-based heading corrections are fused with PDR in an EKF. The resulting

trajectories were compared with compass- and gyro-based PDR baselines and benchmarked against GT using positioning error statistics and spatial overlays.

D. Heading Calibration via Wi-Fi Fingerprinting Results

The heading calibration methodology in Section III-B was evaluated through indoor walking experiments under multiple combinations of $\tau_{\text{var}}^{\text{mag}}$ and $\tau_{\text{rssi}}^{\text{max}}$. For each setting, we computed the mean heading deviation between calibrated and GT headings and counted the total number of calibration events, in order to characterize the tradeoff between calibration accuracy and frequency.

As shown in Fig. 5(a) and (b), stricter thresholds (lower $\tau_{\text{var}}^{\text{wfi}}$ and higher $\tau_{\text{rssi}}^{\text{max}}$) enforce higher spatial consistency and signal quality, which yields lower mean heading errors but fewer accepted calibration events. Relaxing the thresholds increases the number of accepted Wi-Fi solutions, improving coverage along the trajectory, yet also admits less reliable matches and non-line-of-sight (NLOS) observations, which increases the mean heading error. Fig. 5(a) (top) shows that the error grows monotonically as $\tau_{\text{var}}^{\text{wfi}}$ increases and $\tau_{\text{rssi}}^{\text{max}}$ decreases, with the lowest errors (below 5°) obtained under strict settings (e.g., $\tau_{\text{var}}^{\text{wfi}} = 2.0$ m, $\tau_{\text{rssi}}^{\text{max}} = -45$ dBm), at the cost of the fewest calibration events (Fig. 5(a), bottom). In contrast, relaxed settings (e.g., $\tau_{\text{var}}^{\text{wfi}} = 6.0$ m, $\tau_{\text{rssi}}^{\text{max}} = -65$ dBm) maximize calibration count but can exceed 20° error. The 3-D surface in Fig. 5(b) confirms a smooth degradation trend along both thresholds and validates the effectiveness of the proposed dual-filtering strategy.

Fig. 6 illustrates the conditions under which heading calibration was successfully triggered using Wi-Fi fingerprinting. The upper subplot shows the variance (σ_{fpi}^2), while the lower subplot presents the maximum RSS (RSS_{max}) from online scans matched with APs in the offline radio map. Calibration intervals are highlighted as light-blue shaded regions, indicating periods when both quality thresholds were simultaneously satisfied. Each shaded region is bounded by two vertical dotted blue lines: the first marks the scan where a qualified fingerprinting solution was initially detected, and the second indicates the subsequent scan with a second qualified solution. Heading calibration was performed only if additional interscan conditions were also met. For further implementation details, refer to Algorithm 3 in Section III-B5.

Fig. 7 illustrates the impact of the calibrated heading on overall performance. The heading calibrated from

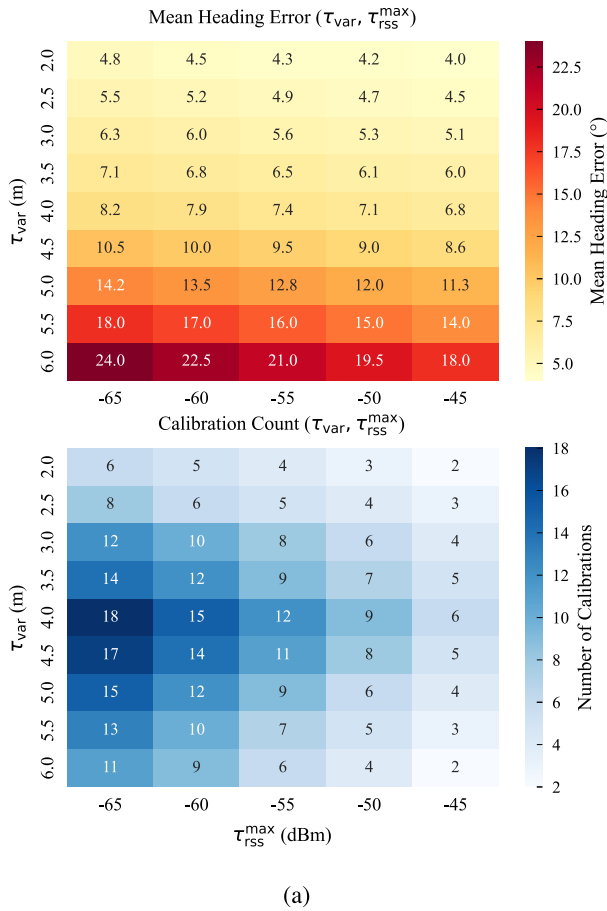


Fig. 5. Visualization of heading calibration performance across different threshold settings of τ_{rss}^{max} and τ_{var}^{wifi} . (a) Heatmaps showing (top) mean heading error and (bottom) number of heading calibrations across varying τ_{rss}^{max} and τ_{var}^{wifi} thresholds. (b) 3-D surface plot of mean heading error with respect to τ_{rss}^{max} and τ_{var}^{wifi} .

high-confidence Wi-Fi fingerprinting solutions (green line) closely follows the GT trajectory (cyan line) and clearly outperforms both the compass and the uncalibrated gyro

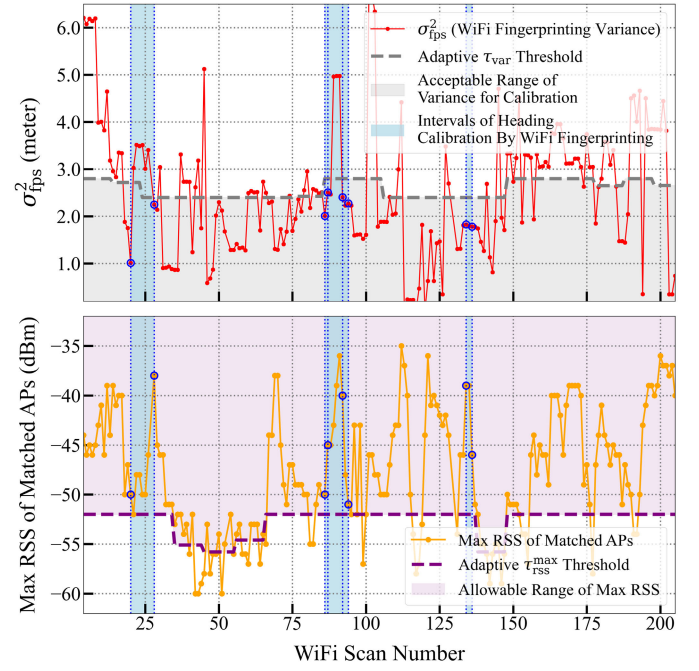


Fig. 6. Heading calibration conditions based on fingerprinting variance and signal strength. The top plot shows fingerprinting variance σ_{fps}^2 with the adaptive threshold τ_{var}^{wifi} ; the shaded region indicates valid calibration intervals. The bottom plot displays the maximum RSS with its threshold τ_{rss}^{max} . Blue-circled markers denote scans triggering calibration, with light-blue shading highlighting valid intervals. Vertical dotted lines mark successful calibration points, with the second being met by additional motion-based conditions.

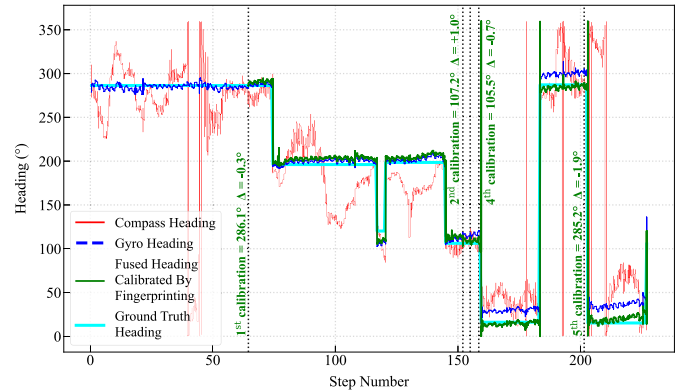


Fig. 7. Comparison of headings calibrated by Wi-Fi fingerprinting, compass, uncalibrated gyroscope, and GT headings. Vertical dotted lines indicate calibration events, annotated with the calibrated heading and its angular deviation (Δ) from the GT.

integration. Fig. 8 compares the heading errors of the compass, gyro, and fused calibrated heading against GT at each step, showing that Wi-Fi-based calibration effectively mitigates drift. After each calibration event, the fused heading error drops sharply, as indicated by the vertical markers annotated with the corrected heading and angular deviation (Δ). The observed deviations of $\Delta = -0.3^\circ$, $+1.0^\circ$, -0.7° , and -1.9° for the 1st, 2nd, 4th, and 5th calibrations confirm the high precision of these corrections.

Fig. 9 presents the empirical cumulative distribution function (ECDF) of heading errors obtained from the compass,

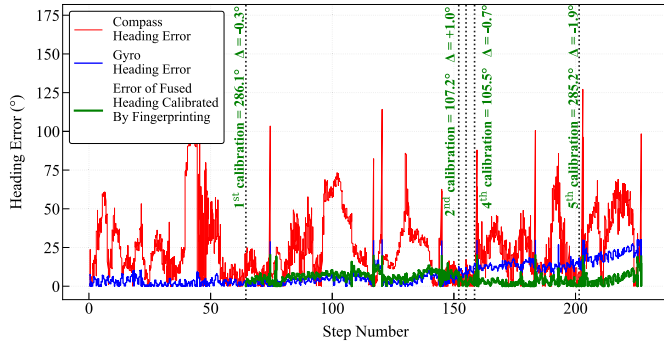


Fig. 8. Heading error comparison before and after calibration using Wi-Fi fingerprinting. The figure illustrates the step-wise heading error (in degrees) for the compass, gyroscope, and calibrated fused heading. Vertical dotted lines indicate the calibration points where the heading was updated using Wi-Fi-based fingerprinting. Each annotation shows the calibrated heading and the signed heading difference (Δ) from the GT.

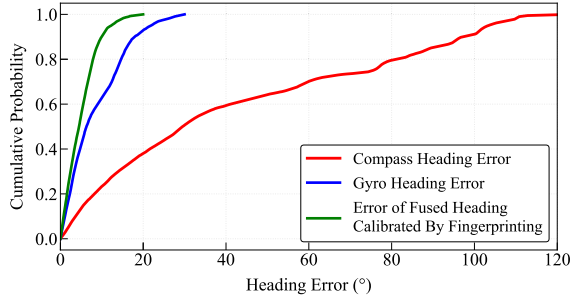


Fig. 9. ECDF of heading error using three sources: compass, gyroscope, and heading calibrated by Wi-Fi fingerprinting.

gyroscope, and Wi-Fi-calibrated fused heading. Quantitatively, the fused heading achieved a median error of 4.50° , a 75th percentile of 7.18° , and a 95th percentile of 12.22° , with a maximum observed error of 19.98° . In contrast, the compass-based heading exhibited a median error of 29.51° and a 95th percentile of 103.32° , while the gyro-only solution yielded a median of 6.30° and a 95th percentile of 21.85° . These results confirm that Wi-Fi-based calibration significantly reduces heading error and enhances robustness against drift.

To evaluate the effectiveness and consistency of the proposed heading calibration method across diverse environments, over 12 controlled experiments were conducted along different paths and in multiple test areas. The results show that 50% of heading errors were below 5.45° , 75% below 8.25° , and 95% below 10.80° . These statistics demonstrate the robustness of the calibration strategy and confirm its practical suitability for improving heading estimation in IPS.

E. Results of Enhanced Heading Calibration Based on QSMFI

1) *Parameter Sensitivity Analysis for Evaluating QSMFI-Based Heading Calibration Under Varying Stability and Sliding-Window Size Parameters:* Fig. 10 illustrates the spatial distributions of the detected QSMFIs and their associated heading error statistics under different combinations of magnetic stability thresholds $\tau_{\text{var}}^{\text{mag}}$ and sliding-window sizes W_s .

The top row of each subfigure presents a 3-D representation of the walking trajectory with QSMFI points overlaid and color-coded by interval index, while the corresponding bottom row depicts heading error distributions across detected intervals using bar plots with accompanying boxplots to summarize error spread and central tendency. As expected, increasing $\tau_{\text{var}}^{\text{mag}}$ leads to a higher number of detected QSMFIs due to a relaxed stability requirement, as seen in Fig. 10(a)–(c). However, this also introduces greater variability in heading accuracy, evident from the broader error bars and more dispersed boxplot statistics in Fig. 10(b) and (c). Notably, Fig. 10(d) compares two configurations with the same $\tau_{\text{var}}^{\text{mag}} = 25$ but different window sizes ($W_s = 250$ versus $W_s = 500$), showing that a larger W_s reduces the total number of accepted QSMFIs but generally improves heading accuracy and consistency, indicating that longer windows enhance the reliability of stability assessment at the expense of reduced detection granularity. These observations support the underlying tradeoff between detection density and heading reliability in quasi-static analysis. Selecting optimal $\tau_{\text{var}}^{\text{mag}}$ and W_s values is therefore crucial and should be guided by the desired balance between temporal resolution and estimation accuracy in downstream positioning or calibration tasks.

Figs. 11 and 12 analyze the sensitivity of heading error and QSMFI detection rate to two key parameters: the sliding-window size W_s and the magnetic variance threshold $\tau_{\text{var}}^{\text{mag}}$. Each subplot reports the cumulative distribution function (cdf) of heading errors across all accepted QSMFIs and includes an inset bar plot indicating the total number of detected intervals for each parameter setting.

In Fig. 11, we fix the magnetic stability threshold $\tau_{\text{var}}^{\text{mag}} \in \{5^\circ, 10^\circ, 20^\circ\}$ and examine the effect of varying W_s . For all $\tau_{\text{var}}^{\text{mag}}$ settings, smaller windows (e.g., $W_s = 100$) lead to a higher number of QSMFIs, over 60 intervals at $\tau_{\text{var}}^{\text{mag}} = 5^\circ$, due to the shorter observation period for variance estimation. However, this comes with reduced heading accuracy, with median errors ranging from 10° to 15° , and long-tailed error distributions that degrade the reliability of the calibration. As W_s increases to 300, the number of detected intervals drops significantly (fewer than 20 for $\tau_{\text{var}}^{\text{mag}} = 5^\circ$), but the error distributions become more concentrated: the 90th percentile heading error falls below 15° for all curves, indicating greater consistency and suppression of noisy intervals.

Conversely, Fig. 12 fixes $W_s \in \{100, 200, 300\}$ and explores the effect of increasing $\tau_{\text{var}}^{\text{mag}} \in \{5^\circ, 10^\circ, 15^\circ, 20^\circ\}$. Here, higher thresholds permit more intervals by relaxing the stability criterion; for instance, more than 70 intervals are detected at $\tau_{\text{var}}^{\text{mag}} = 20^\circ$ for $W_s = 100$, but with a substantial increase in heading error spread. Notably, the CDFs for $\tau_{\text{var}}^{\text{mag}} = 20^\circ$ become flatter, especially for smaller W_s , indicating degraded precision. In contrast, lower thresholds (e.g., $\tau_{\text{var}}^{\text{mag}} = 5^\circ$) enforce stricter stability, yielding fewer but more accurate intervals, with median heading errors often below 10° for $W_s = 200$ and 300.

Overall, these results highlight a fundamental tradeoff: aggressive parameter settings (small W_s , high $\tau_{\text{var}}^{\text{mag}}$) boost QSMFI density but compromise heading accuracy, whereas conservative configurations reduce interval frequency but yield

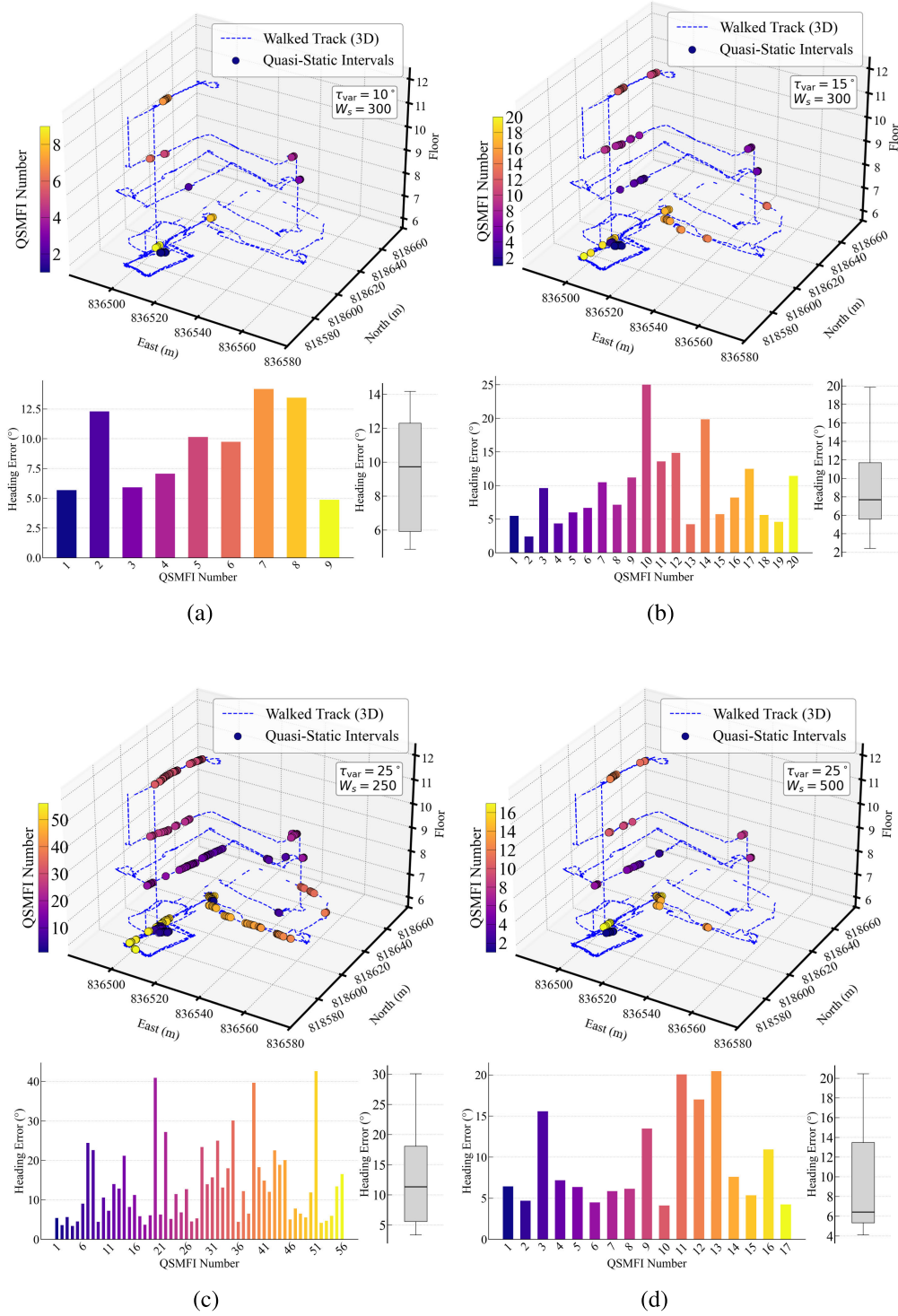


Fig. 10. Location distributions of detected QSMFIs (top of each subfigure) and corresponding heading error statistics (bottom) under varying magnetic stability thresholds $\tau_{\text{var}}^{\text{mag}}$ and sliding-window size W_s . (a) $\tau_{\text{var}}^{\text{mag}} = 10^\circ$, $W_s = 300$. (b) $\tau_{\text{var}}^{\text{mag}} = 15^\circ$, $W_s = 300$. (c) $\tau_{\text{var}}^{\text{mag}} = 25^\circ$, $W_s = 250$. (d) $\tau_{\text{var}}^{\text{mag}} = 25^\circ$, $W_s = 500$.

more reliable calibration. For applications that prioritize heading fidelity, a moderate setting (e.g., $W_s = 200$, $\tau_{\text{var}}^{\text{mag}} = 10^\circ$) appears to offer a favorable balance, producing over 40 intervals with median errors near 8° .

2) *Impact of Adaptive Sliding-Window Size and Variance Threshold:* This section evaluates the performance and operational benefits of the proposed adaptive mechanism for

selecting the sliding-window size W_s and magnetic variance threshold $\tau_{\text{var}}^{\text{mag}}$ used in QSMFI detection. The aim is to assess how dynamically tuning these parameters, based on motion context and temporal heading consistency, can improve heading calibration accuracy and increase the number of valid QSMFIs detected in real time. We compare the adaptive approach to a baseline method that relies on fixed values for

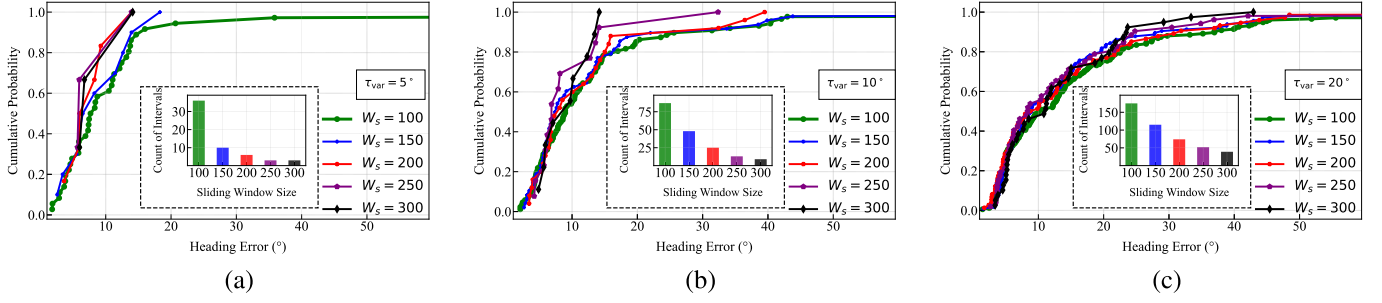


Fig. 11. Effect of varying sliding-window size $W_s \in \{100, 200, 300\}$ on heading error estimation for fixed magnetic variance thresholds $\tau_{\text{var}}^{\text{mag}} \in \{5^\circ, 10^\circ, 20^\circ\}$, evaluated at a sampling frequency of 50 Hz. Each subplot presents CDFs of heading error and an inset showing the number of detected QSMFIs for each window size. (a) $\tau_{\text{var}}^{\text{mag}} = 5^\circ$. (b) $\tau_{\text{var}}^{\text{mag}} = 10^\circ$. (c) $\tau_{\text{var}}^{\text{mag}} = 20^\circ$.

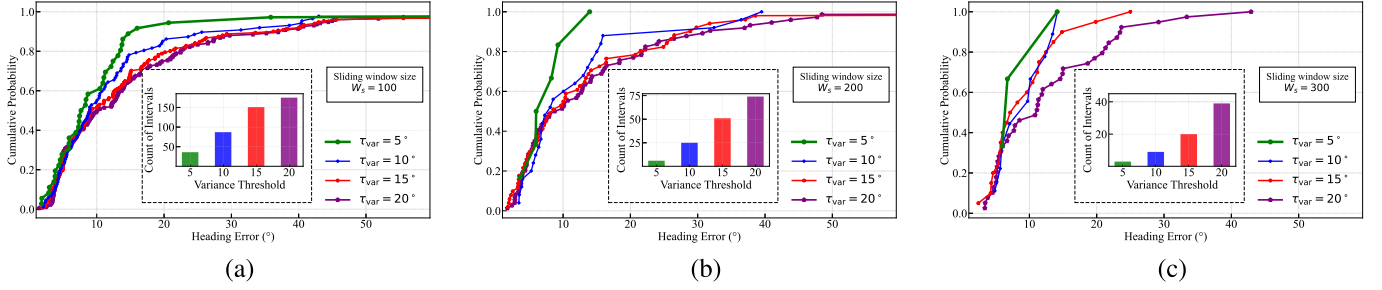


Fig. 12. Effect of varying magnetic variance threshold $\tau_{\text{var}}^{\text{mag}} \in \{5^\circ, 10^\circ, 15^\circ, 20^\circ\}$ on heading error estimation for fixed sliding-window sizes $W_s \in \{100, 200, 300\}$, evaluated at a sampling frequency of 50 Hz. Each subplot presents CDFs of heading error and an inset showing the number of detected QSMFIs for each threshold. (a) $W_s = 100$. (b) $W_s = 200$. (c) $W_s = 300$.

W_s and $\tau_{\text{var}}^{\text{mag}}$, and examine how initialization with conservative settings transitions into context-aware adaptation. The analysis draws on the sensitivity results established in Section IV-E1 and the methodological framework described in Section III-C.

At the initial stage of system operation, the algorithm intentionally employed a conservative strategy by selecting relatively large sliding-window sizes W_s and strict variance thresholds $\tau_{\text{var}}^{\text{mag}}$. This initialization was based on outcomes inferred from the sensitivity analysis in Section IV-E1, which demonstrated that such settings yield more stable and accurate QSMFI detections. The goal was to capture only the most robust and trustworthy QSMFIs early on, ensuring that the heading references used to evaluate the consistency of future intervals were themselves highly reliable.

These high-confidence initial QSMFIs serve as anchors to bootstrap the adaptive logic. As user motion continued and more sensor observations accumulated, the detection parameters were gradually adapted to the dynamic motion context, as outlined in Section III-C. During walking, the system adjusted W_s and $\tau_{\text{var}}^{\text{mag}}$ in response to real-time conditions such as heading variability and gyroscope-compass alignment. When stable QSMFIs were repeatedly validated, the thresholds were cautiously adjusted to maintain both detection frequency and calibration accuracy. Conversely, when valid calibration intervals became rare, the system relaxed the constraints to enhance the likelihood of detection without compromising reliability. To maintain calibration integrity, all adaptive updates were bounded within predefined operating ranges inferred from the sensitivity analysis. This bounded adaptation ensured that W_s and $\tau_{\text{var}}^{\text{mag}}$ never deviated into regimes likely to produce unreliable or noisy heading estimates, preserving system robustness.

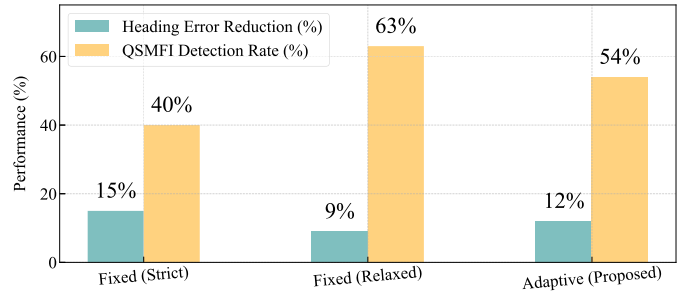


Fig. 13. Comparison of heading error reduction and QSMFI detection rate between fixed and adaptive strategies.

The benefit of this adaptive framework was quantitatively assessed by quantifying the performance benefits of the proposed adaptive calibration strategy. The evaluation compares three strategies: 1) fixed parameters with strict thresholds; 2) fixed parameters with relaxed thresholds; and 3) the proposed adaptive approach. Fig. 13 presents a comparative visualization of the two key performance metrics: heading error reduction and QSMFI detection rate. The heading error reduction percentage was computed by comparing the mean heading error under each strategy to the baseline gyroscope-only heading error estimated from the GT heading. Specifically, the mean heading error for each method was divided by the gyroscope error baseline, and the resulting improvement was expressed as a percentage. The QSMFI detection rate was computed as the ratio of accepted QSMFI intervals to the total duration of the trajectory period. The results indicate that the adaptive method achieves a

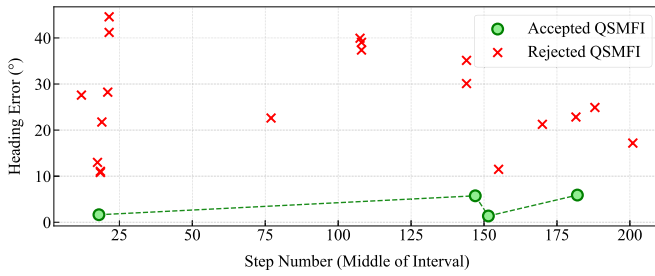


Fig. 14. Heading error of accepted and rejected QSMFIs by self-attention-based fault rejection mechanism.

12% reduction in heading error relative to the gyroscope baseline, while the fixed strategies achieve 15% (strict) and 9% (relaxed), respectively. Notably, although the fixed strict strategy provides slightly higher error reduction, it does so at the cost of a significantly lower detection rate (40%). In contrast, the adaptive method maintains a moderate error reduction benefit while achieving a 54% detection rate, closer to the relaxed configuration's 63%, but with improved reliability.

3) *Validation of Self-Attention and Rejection of Faulty QSMFIs*: The effectiveness of the self-attention-based QSMFI validation method (see Section III-C4) is illustrated in Fig. 14, which plots the heading error of each detected QSMFI against its step number. Accepted intervals (green circles) consistently show low errors, typically under 6° , while rejected ones (red crosses) exhibit significantly higher errors, with many exceeding 40° . This clear separation underscores the reliability of the proposed cross correlation consistency check, which compares magnetometer-derived heading changes across QSMFI boundaries with gyroscope-integrated rotations. Intervals are rejected when the discrepancy surpasses the τ_Δ threshold. The results confirm that accepted intervals correspond to consistent and stable heading estimates, while rejected ones often arise in magnetically unstable or distorted regions. Importantly, accepted QSMFIs are temporally and spatially distributed along the trajectory, ensuring calibration opportunities without excessive rejection. This validates the proposed strategy as a lightweight and effective mechanism for suppressing faulty QSMFIs, enhancing the accuracy and robustness of heading calibration across diverse walking conditions.

4) *Evaluation of Heading Accuracy With QSMFI Calibration*: The effectiveness of the proposed QSMFI-based calibration framework is assessed by comparing the heading error distributions of three sources: compass, gyroscope integration, and the QSMFI-calibrated heading estimates. Fig. 15 illustrates the temporal behavior of heading estimates over step sequences, highlighting QSMFI-based calibration points, while Fig. 16 presents the ECDFs of heading errors.

The ECDF results clearly demonstrate the substantial improvement achieved through QSMFI-based calibration. The raw compass heading suffers from large and highly variable errors, with a mean of 37.13° and a maximum deviation exceeding 173° . In contrast, the gyroscope-based heading,

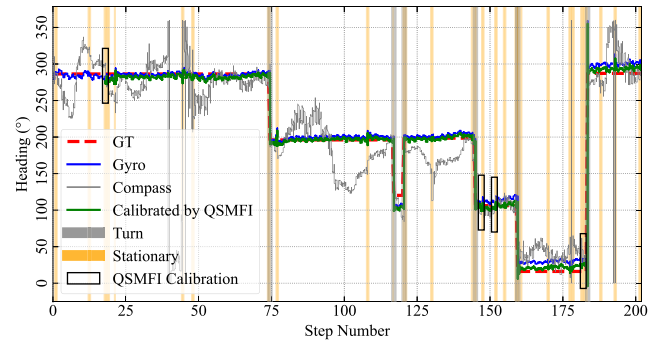


Fig. 15. Comparison between step-wise heading estimates from the GT, gyroscope, compass, and QSMFI-calibrated estimates. Black rectangles denote calibration points where QSMFIs were accepted and applied for heading correction.

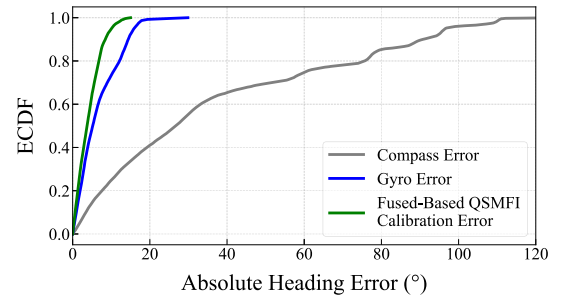


Fig. 16. ECDF of absolute heading errors for the compass (gray), gyroscope (blue), and QSMFI-calibrated heading (green). The QSMFI-calibrated estimate demonstrates significantly improved accuracy and reduced error variability.

though generally more stable, still exhibits accumulated drift, resulting in a mean error of 6.62° . The proposed QSMFI-calibrated heading substantially outperforms both sources, with a reduced mean error of 4.19° and median error of only 3.74° . The standard deviation is also minimized (3.05°), and the maximum error is bounded below 15° .

F. Comparison Between Different Positioning Solutions

Fig. 17 illustrates the practical impact of heading calibration on overall PDR performance and trajectory alignment, compared to those derived from uncalibrated gyroscope and compass headings. Notably, the trajectory derived from the gyroscope heading exhibits significant drift, particularly toward the end of the walking path or overextended walking intervals, reflecting the cumulative effect of gyro bias. In contrast, the trajectory generated using the calibrated heading closely aligns with the interpolated GT path, offering clear improvements in walking direction and localization accuracy. This outcome highlights the ability of the proposed calibration method to effectively mitigate gyro drift and maintain trajectory reliability over long walking sequences, surpassing both compass- and gyro-only approaches.

Fig. 18 illustrates the trajectory comparison between GT, PDR, and the EKF-based fusion of PDR and Wi-Fi fingerprinting, where the PDR solution benefits from heading correction derived through fingerprinting and QSMFIs. The EKF trajectory further enhances accuracy by integrating PDR

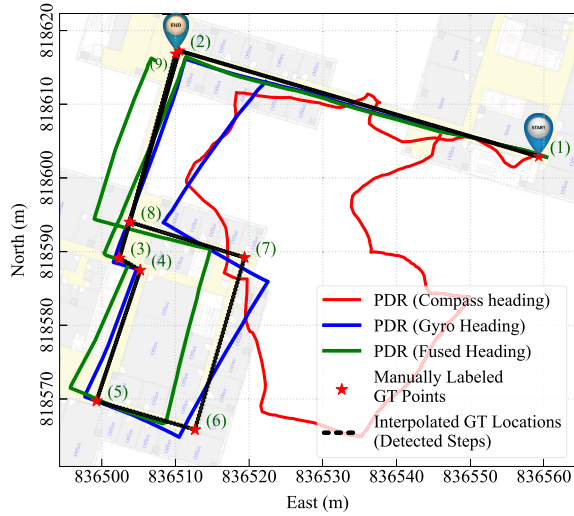


Fig. 17. Evaluation of the performance of PDR using compass, gyro, and fused headings calibrated by Wi-Fi fingerprinting and QSMFIs.

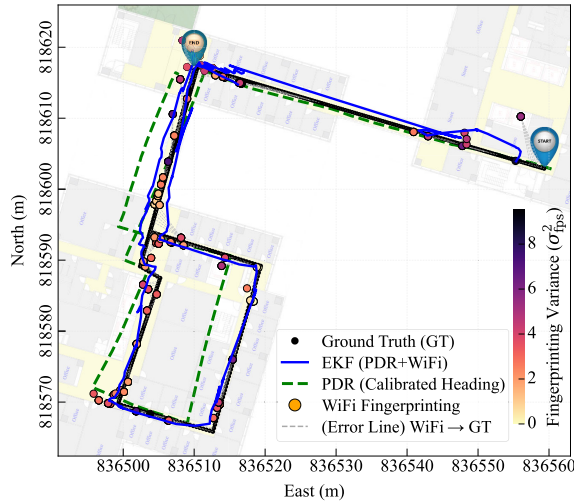


Fig. 18. Trajectory comparison of PDR with calibrated heading by Wi-Fi fingerprinting and QSMFIs, EKF, and Wi-Fi fingerprinting.

and fingerprinting, compensating for drift and spatial noise. Wi-Fi position estimates are visualized using color-coded markers, where the color scale denotes the variance in fingerprinting confidence (σ_{fps}^2). The dashed gray connectors from each Wi-Fi estimate to its corresponding GT location reveal the magnitude and direction of positioning error. Overall, the fusion approach demonstrates superior performance and reduced positioning error.

Fig. 19 presents a comparative evaluation of location error distributions for five positioning solutions. The compass-based PDR exhibits the highest error variability and median error, confirming its susceptibility to magnetic distortions in indoor environments. Replacing the compass with a gyroscope significantly improves performance, with lower error spread and reduced outliers. Further gains are achieved by calibrating the heading, which yields a tighter interquartile range and a lower median, indicating more consistent performance. Wi-Fi fingerprinting shows competitive accu-

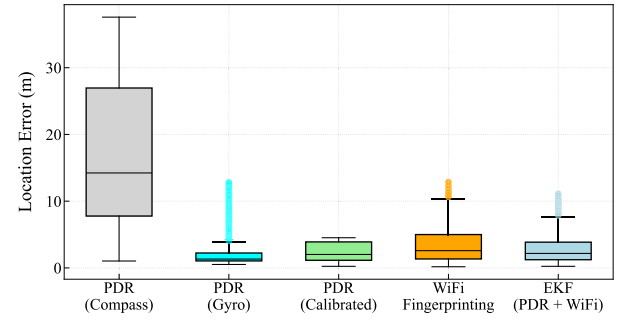


Fig. 19. Boxplot comparison of location errors across different positioning methods.

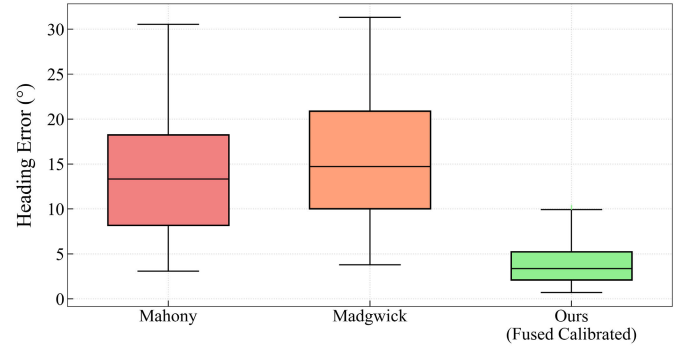


Fig. 20. Comparative assessment of different heading estimation methods.

racy, with a moderate error profile and fewer extreme values. The EKF fusion method, integrating PDR and Wi-Fi, achieves the best overall performance, combining low median error with minimal variance. These results underscore the value of heading refinement and multisource data fusion for enhancing indoor localization robustness and precision.

G. Comparative Assessment and Discussion

1) *Comparative Assessment of Heading Methods:* To validate the effectiveness of the proposed heading calibration framework, we conducted a comparative assessment against two baseline methods (Mahony complementary filter (CF) [35] and Madgwick gradient descent filter [36]) in addition to the compass-only heading and gyroscope-based heading that were discussed above in Fig. 16. The Mahony and Madgwick filters represent widely adopted sensor fusion approaches in attitude and heading reference systems (AHRSs) [37]. Both filters fuse gyroscope, accelerometer, and magnetometer measurements to estimate orientation, with Mahony employing a proportional-integral (PI) controller and Madgwick using gradient descent optimization. While these CFs excel at short-term sensor fusion and noise suppression, they inherently lack external absolute references for long-term drift correction, relying solely on the magnetometer for heading stabilization.

Fig. 20 presents the distribution of heading errors across all five methods evaluated on the same trajectory dataset. The Mahony and Madgwick filters achieve mean heading errors of 14.0° and 15.8° , respectively, demonstrating their

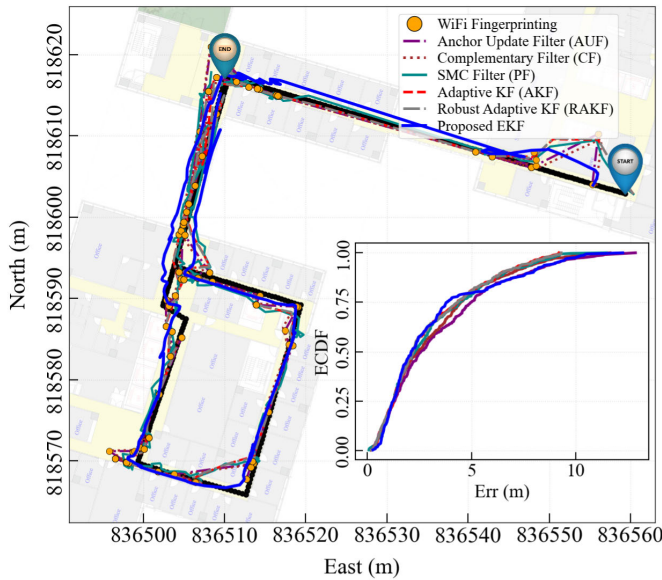


Fig. 21. Comparative assessment of different fusion methods.

capability to reduce gyroscope drift compared to gyroscope-only estimates (9.4°). However, both filters remain susceptible to magnetic disturbances and systematic biases inherent in smartphone magnetometers, particularly in indoor environments with pervasive ferromagnetic interference. In contrast, the proposed fused calibrated heading achieves a mean error of 4.0° , representing substantial improvements of 71.6% over Mahony and 74.8% over Madgwick. The narrow interquartile range (2.1° to 5.2°) further demonstrates the robustness of the proposed approach under varying motion patterns and environmental conditions. This superior performance stems from the integration of two complementary calibration sources: Wi-Fi fingerprinting-based heading corrections that provide infrastructure-assisted absolute references, and QSMFI-based magnetometer calibration that selectively exploits magnetically stable intervals while rejecting disturbed measurements. Unlike CFs that continuously integrate potentially corrupted magnetometer readings, the proposed method adaptively qualifies and applies heading corrections only when reliability conditions are satisfied, thereby achieving both accuracy and robustness.

2) *Comparative Assessment of Fusion Methods:* To evaluate the effectiveness of the proposed loosely coupled EKF framework with adaptive heading calibration, we compared it against five state-of-the-art Wi-Fi/PDR fusion methods on the same test trajectory. The compared methods include: 1) anchor update filter (AUF), which resets the PDR trajectory whenever accurate Wi-Fi fix is available and propagates dead reckoning in between, similar in spirit to anchor- and map-assisted Wi-Fi/PDR fusion in [38]; 2) CF, which combines inertial and Wi-Fi measurements through weighting factor; 3) sequential Monte Carlo filter (PF), which uses particle filtering for non-Gaussian noise handling [39], [40]; 4) adaptive Kalman filter (AKF), which dynamically adjusts measurement noise covariance in line with AKF-based indoor positioning methods in [38], [41], and [42]; and 5) robust AKF (RAKF),

which incorporates t-test-based outlier detection for enhanced robustness [43]. All methods were configured to utilize the same calibrated heading estimates derived from the proposed Wi-Fi fingerprinting and QSMFI detection modules, ensuring that observed performance differences stem primarily from the fusion strategy rather than heading quality. Fig. 21 presents both the 2-D trajectory overlay and the ECDF of positioning errors. The trajectory visualization demonstrates that all fusion methods closely follow the GT path, with the proposed EKF maintaining smooth and consistent tracking throughout the test route. The ECDF curves reveal that the proposed method achieves competitive performance across all error percentiles, with particularly strong performance in the 50th to 90th percentile range. While the particle filter exhibits slightly lower median error due to its ability to handle multimodal distributions, the proposed EKF offers comparable accuracy with significantly lower computational complexity, making it more suitable for real-time smartphone-based navigation. The adaptive methods (AKF and RAKF) demonstrate similar performance to the proposed approach, confirming that adaptive measurement weighting is beneficial. However, the proposed method's integration of heading calibration with adaptive thresholding provides a more holistic solution that addresses both heading drift and position uncertainty. Overall, the results validate that the proposed loosely coupled EKF framework, when combined with the adaptive heading calibration strategy, delivers robust and accurate positioning performance that is competitive with or superior to existing state-of-the-art fusion methods.

V. CONCLUSION

This article presents a robust and adaptive heading calibration framework for smartphone-based indoor positioning, integrating Wi-Fi fingerprinting and QSMFI detection. The proposed method dynamically evaluates fingerprinting calibration opportunities using spatial dispersion and signal-strength-based adaptive thresholding informed by self-learning. Experimental results demonstrate that the Wi-Fi-based heading calibration reduces the median heading error by approximately 83% compared to compass-based methods and by about 17% compared to gyro-based methods. In addition, the adaptive QSMFI scheme achieves approximately 89% improvement in mean heading accuracy compared to compass estimates, and about 43% improvement over gyro-only heading. Integrating these calibrated headings within a tightly coupled EKF further enhances positioning accuracy and stability, significantly outperforming traditional standalone methods. These quantitative improvements confirm the effectiveness of the proposed system in reducing drift and enhancing robustness. Future work may explore the application of deep reinforcement learning to optimize detection thresholds and sliding-window parameters based on contextual data, enabling more effective exploitation of QSMFIs under varying environmental and motion conditions. Another promising direction is the development of personalized calibration strategies through user-specific modeling of motion dynamics and smartphone handling behaviors, which can further enhance the reliability and adaptability of the heading calibration framework.

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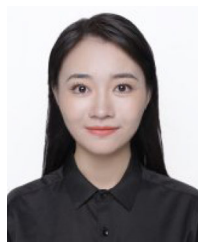
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